ISI Paper Series — Paper 2

Empirical Results and Distributional Structure of the International Sovereignty Index (ISI) EU-27, Vintage 2024, Methodology v1.0

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Abstract

This paper reports the first empirical application of the International Sovereignty Index (ISI) to the 27 member states of the European Union using vintage-2024 data under methodology v1.0. The ISI composite, defined as the unweighted arithmetic mean of six normalised axes (financial, energy, technology, defence, critical inputs, logistics), ranges from 0.23567126 (France, rank 27) to 0.51748148 (Malta, rank 1) with a population mean of 0.34439144 and population standard deviation of 0.06992188. One country is classified as *highly concentrated*, 23 as *moderately concentrated*, and three as *mildly concentrated*. A variance decomposition shows that defence (35.1%) and logistics (34.3%) together account for 69.4% of cross-country composite variance. Leave-one-axis-out analysis confirms that removing the defence axis produces the largest rank disruption (Spearman $\rho = 0.833$). Dirichlet

weight perturbation (10 000 draws, $\alpha_j = 10$) yields a mean rank-order correlation of $\rho = 0.979$, with the Netherlands exhibiting the highest rank volatility ($\sigma = 3.07$). Z-score standardisation produces $\rho = 0.896$ against the baseline, indicating moderate sensitivity to implicit axis scaling. Winsorisation of the two highest-variance axes at the 95th percentile preserves the full rank order ($\rho = 1.000$). The paper provides radar profiles for eight selected countries, full axis-level rankings, and a complete reproducibility protocol.

JEL codes: F02, F15, F52, O30, Q43 *[to confirm]*

Keywords: sovereignty index, EU-27, concentration measurement, composite indicator, supply-chain dependence, defence dependence

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Executive Summary

This paper presents the complete empirical results of the International Sovereignty Index (ISI) applied to the 27 member states of the European Union, using vintage-2024 data processed under methodology v1.0. The index measures the concentration of bilateral external dependencies across six structural axes. Higher scores indicate higher concentration of counterparty exposure. The principal findings are:

- **Compressed composite distribution.** The ISI composite ranges from 0.23567126 (France) to 0.51748148 (Malta), a span of 0.28181022 on the unit interval. The population standard deviation is 0.06992188, yielding a coefficient of variation of 0.2030. The interquartile range is 0.09881618 (P25 = 0.29179051, P75 = 0.39060669). See Table 3 and Figure 1.

- **Tier compression.** Under the four-tier classification, 23 of 27 countries (85.2%) fall in the *moderately concentrated* tier (ISI $\in [0.25, 0.50)$). Only Malta exceeds 0.50 (*highly concentrated*); three countries (Latvia, Slovakia, France) fall below 0.25 (*mildly concentrated*). No country is classified as *unconcentrated*. See Table 5 and Figure 4.

- **Defence and logistics dominate cross-country variance.** A covariance-based variance decomposition attributes 35.1% of composite variance to the defence axis and 34.3% to the logistics axis. Together, these two axes account for 69.4% of total composite variance. Financial concentration contributes only 1.0%. See Table 10 and Figure 6.

- **Axis dominance is concentrated.** In 16 of 27 countries, the defence axis records the highest axis-level score; in the remaining 11, logistics is highest. No country has its maximum score on the financial, energy, technology, or critical-inputs axis. See Table 26.

- **Three moderate cross-axis correlations.** Of 15 pairwise Pearson correlations, three exceed $|r| \geq 0.30$: critical inputs–logistics ($r = 0.549$), energy–technology ($r = 0.530$), and energy–critical inputs ($r = 0.372$). The remaining 12 pairs are weak ($|r| < 0.30$), indicating that the six axes capture largely distinct dimensions. See Table 11 and Figures 7 and 8.

- **Robustness to weight perturbation.** Under Dirichlet weight perturbation (10 000 Monte Carlo draws, $\alpha_j = 10$), the mean Spearman rank correlation with the baseline is $\rho = 0.979$. The Netherlands exhibits the highest rank volatility (standard deviation 3.07 ranks; 90% band: 10–20). See Table 16.

- **Sensitivity to defence axis removal.** Leave-one-axis-out analysis yields Spearman correlations ranging from $\rho = 0.833$ (defence excluded) to $\rho = 0.993$ (financial

excluded). Removing the defence axis produces the largest rank disruptions; Slovakia moves from rank 26 to rank 27, and Malta from rank 1 to rank 2, when defence is excluded. See Tables 14 and 27.

- **Z-score standardisation alters rankings materially.** When each axis is z-scored before averaging, the rank-order correlation with the baseline drops to $\rho = 0.896$, indicating moderate sensitivity to the implicit weighting induced by axis dispersion. See Table 15.

- **Winsorisation preserves full rank order.** Capping the defence and logistics axes at their respective 95th percentiles and recomputing produces identical rank ordering ($\rho = 1.000$), indicating that extreme axis-level values do not drive rank inversion. See Table 17.

- **Slovakia's zero defence score.** Slovakia records a defence axis score of 0.00000000, the only zero score in the dataset. This arises mechanically from the absence of bilateral supplier entries for Slovakia in the SIPRI arms-transfer data under methodology v1.0. It does not imply the absence of defence procurement. See Section 7.

1 Introduction and Contribution

1.1 Motivation

Composite indicators that measure structural features of national economies have a long tradition in institutional research. Indices published by the OECD, the World Economic Forum, and various United Nations agencies compress multidimensional data into scalar summaries that facilitate cross-country comparison and policy monitoring [1]. The International Sovereignty Index (ISI) contributes to this tradition by focusing specifically on *concentration of bilateral external dependencies*: the degree to which a country's cross-border economic and strategic linkages are concentrated among a small number of counterparty states.

The ISI does not measure sovereignty in a political or legal sense. Rather, it quantifies a structural feature—counterparty concentration—that is a prerequisite for informed policy analysis of external dependence. A high ISI score indicates that a country's external flows in a given domain are heavily concentrated among few partners; a low score indicates dispersion across many partners. The index makes no normative claim about whether concentration is advantageous or disadvantageous.

1.2 Scope of this paper

This paper—the second in the ISI series—presents the full empirical results of the first ISI application: 27 EU member states, vintage-2024 data, methodology v1.0. The methodological foundations (axis definitions, normalisation procedures, data sources, aggregation rule, integrity chain) are documented in Paper 1 [2]. The present paper is self-contained for readers interested in the empirical findings; cross-references to Paper 1 are provided where construction details are relevant to interpretation.

The specific contributions of this paper are:

1. A complete composite and axis-level ranking of all 27 EU member states with eight-decimal precision, accompanied by classification labels.

2. A distributional analysis of the composite index, including summary statistics, percentiles, a Lorenz curve, and a Gini coefficient.

3. An axis-level diagnostic framework that identifies which axes drive cross-country variation, using a covariance-based variance decomposition and axis-dominance analysis.

4. A cross-axis correlation analysis identifying moderate pairwise associations and representing them in a network graph.

5. A four-part robustness programme: leave-one-axis-out (LOO) analysis, z-score standardisation, Dirichlet weight perturbation (Monte Carlo), and winsorisation of high-variance axes.

6. Radar-chart profiles for eight selected countries illustrating the heterogeneity of axis-level patterns.

7. A reproducibility protocol and computational appendix enabling full replication from the provided snapshot.

1.3 Interpretation conventions

Throughout this paper, the following conventions apply:

- Higher ISI composite ⇒ higher concentration ⇒ greater structural dependence on a small set of counterparties.
- Rank 1 = most concentrated country; rank 27 = least concentrated.
- The terms "most concentrated" and "least concentrated" are descriptive. They are not synonymous with "worst" and "best."
- Correlation between axes does not imply causation. Where plausible structural interpretations exist, they are stated as hypotheses, not conclusions.
- Population standard deviation (N denominator) is used throughout because the 27 EU member states constitute the full target population, not a sample drawn from a larger universe.

1.4 Structure

The remainder of this paper is organised as follows. Section 2 summarises the data sources, aggregation rule, and classification thresholds. Section 3 presents the composite distribution, full ranking, and tier-compression analysis. Section 4 provides axis-level diagnostics including dispersion, dominance, and variance decomposition. Section 5 examines the pairwise correlation structure. Section 6 reports four robustness and sensitivity exercises. Section 7 presents radar-chart profiles for eight selected countries. Section 8 discusses interpretation boundaries and limitations. Section 9 concludes. Appendices A–D provide full tables, robustness details, the reproducibility protocol, and the computational code listing.

2 Data and Construction

This section provides a concise overview of the ISI's data inputs, aggregation rule, and classification thresholds. Full construction details—including normalisation procedures, missing-data protocols, fallback basis rules, and the integrity chain—are documented in Paper 1 [2].

2.1 Scope and vintage

The results in this paper are computed from the ISI v1.0 snapshot for vintage year 2024. The scope covers $N = 27$ EU member states. All 27 countries have complete data across all six axes (`complete = true`).

2.2 Axes and data windows

The ISI comprises six axes, each measuring counterparty concentration in a distinct domain of cross-border exposure. Table 1 summarises each axis, its primary data source, and the observation window used for the 2024 vintage.

Table 1: ISI axes: definitions, primary data sources, and observation windows for vintage 2024.

Axis	Domain	Primary Source	Window
1	Financial sovereignty	BIS LBS, IMF CPIS [3, 4]	2022–2024
2	Energy dependency	IEA, Eurostat [5, 6]	2022–2024
3	Technology / semiconductor dependency	Eurostat Comext [6]	2022–2024
4	Defence industrial dependency	SIPRI TIV [7]	2019–2024
5	Critical inputs / raw materials	UN Comtrade, Eurostat [6, 8]	2022–2024
6	Logistics / freight dependency	Eurostat Comext [6]	2022–2024

Note. The defence axis uses a six-year rolling window (2019–2024) rather than the three-year window applied to other axes. This reflects the lower frequency and higher lumpiness of arms-transfer flows, as documented in Paper 1 [2].

2.3 Aggregation rule

The ISI composite for country i is the unweighted arithmetic mean of the six axis-level scores:

$$\mathrm{ISI}_i = \frac{1}{6}\sum_{j=1}^{6} A_{j,i}\,, \qquad A_{j,i} \in [0,\,1]\,, \tag{1}$$

where $A_{j,i}$ denotes the normalised score of country i on axis j. Each axis score is bounded to the unit interval by the normalisation procedure defined in Paper 1.

The equal-weight specification is the baseline. Section 6 examines sensitivity to alternative weighting schemes through Dirichlet perturbation.

2.4 Classification thresholds

Countries are classified into four tiers based on the composite score:

Table 2: ISI classification tiers.

Tier	**Composite range**	**Count (EU-27)**
Highly concentrated	ISI ≥ 0.50	1 (Malta)
Moderately concentrated	$0.25 \leq$ ISI < 0.50	23
Mildly concentrated	$0.15 \leq$ ISI < 0.25	3 (Latvia, Slovakia, France)
Unconcentrated	ISI < 0.15	0

The thresholds are structurally analogous to the Herfindahl–Hirschman Index (HHI) tiers used in antitrust analysis (see [9]), adapted to the ISI's unit-interval scale. The threshold values are defined in methodology v1.0 and documented in Paper 1.

2.5 Rounding and precision

All axis-level and composite scores are reported to eight decimal places, matching the precision of the authoritative ISI snapshot. Population-level statistics (mean, standard deviation, percentiles) are computed with full floating-point precision and reported to eight decimal places unless otherwise noted. The population standard deviation (σ, with N denominator) is used throughout, as the 27 EU member states constitute the complete target population.

3 Composite Results: Distribution, Ranking, and Tier Compression

3.1 Summary statistics

Table 3 reports the distributional summary of the ISI composite across the 27 EU member states.

Table 3: ISI composite: summary statistics, EU-27, 2024.

Statistic	**Value**
N	27.00000000
Minimum	0.23567126
Maximum	0.51748148
Range	0.28181022
Mean	0.34439144
Median (P50)	0.34027984
Std. dev. (pop.)	0.06992188
Std. dev. (sample)	0.07125384
Coeff. of variation (pop.)	0.20303651

The composite mean (0.34439144) slightly exceeds the median (0.34027984), indicating a mild positive skew driven by Malta's outlying score of 0.51748148. The coefficient of variation (0.203) is relatively low, suggesting that the composite scores are compressed relative to the full unit-interval range.

3.2 Percentiles

Table 4: ISI composite: selected percentiles.

Percentile	**Value**
P10	0.25534441
P25	0.29179051
P50	0.34027984
P75	0.39060669
P90	0.42586650

The interquartile range (IQR = P75 − P25 = 0.09881618) spans approximately 9.9 percentage points of the unit interval. The P10–P90 range is 0.17052209, indicating that the central 80% of the distribution occupies roughly 17 percentage points.

3.3 Composite distribution

Figure 1 presents a histogram of the composite scores. The distribution is unimodal with its mode in the 0.30–0.35 bin. A visible right tail is produced by Malta and Cyprus.

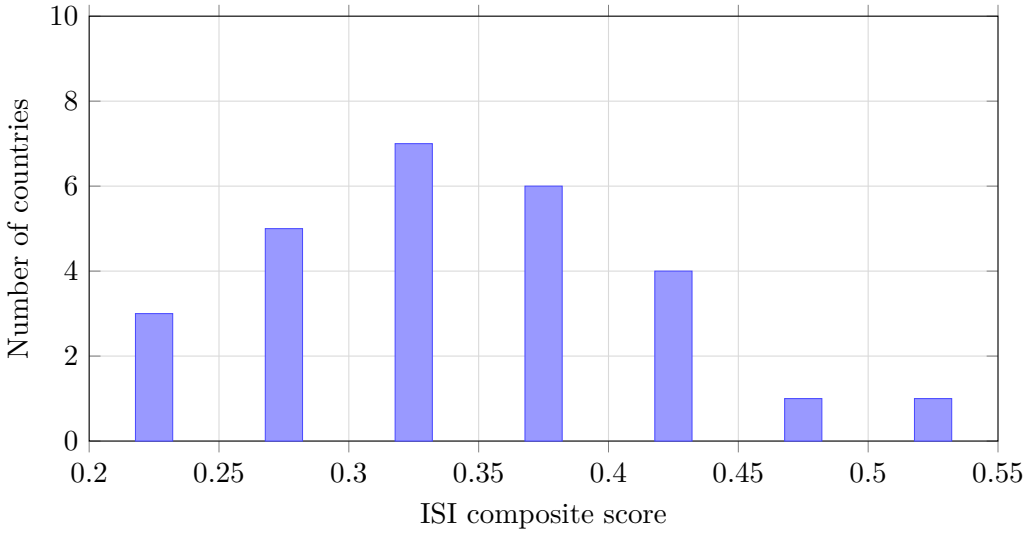

Figure 1: Histogram of ISI composite scores, EU-27, 2024. Bin width = 0.05. The dashed vertical line marks the population mean (0.34439144).

Figure 2 shows the empirical cumulative distribution function (ECDF), which rises steeply between 0.26 and 0.42 and flattens at the tails, confirming the compressed mid-range clustering.

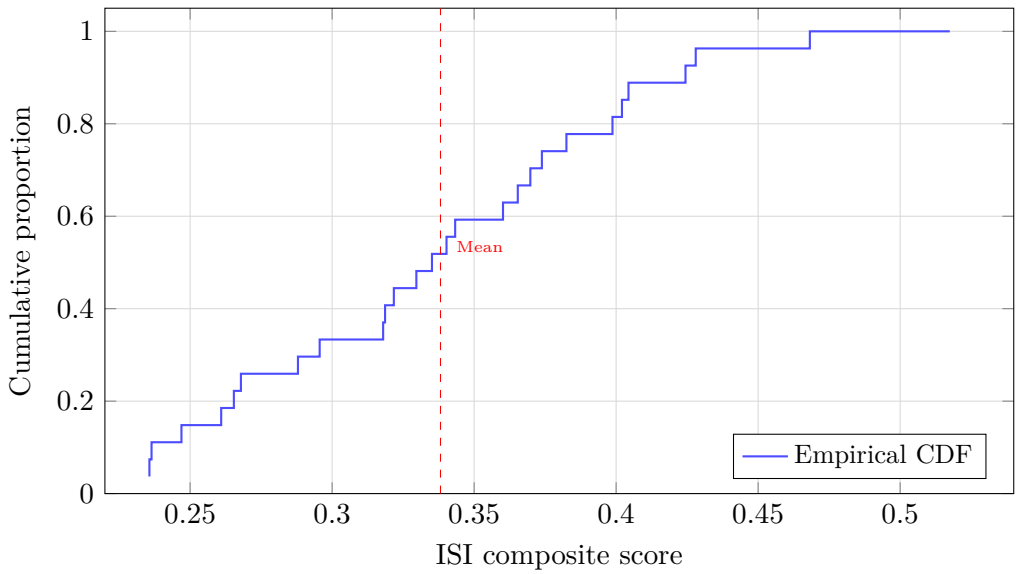

Figure 2: Empirical cumulative distribution function (ECDF) of ISI composite scores, EU-27, 2024.

3.4 Lorenz curve and Gini coefficient

To further characterise the distribution, Figure 3 plots the Lorenz curve of the composite scores (countries ordered from lowest to highest ISI, cumulative share of total ISI on the vertical axis). The Gini coefficient is 0.1155, indicating a relatively egalitarian distribution of composite scores across countries. This is consistent with the tier compression observed above.

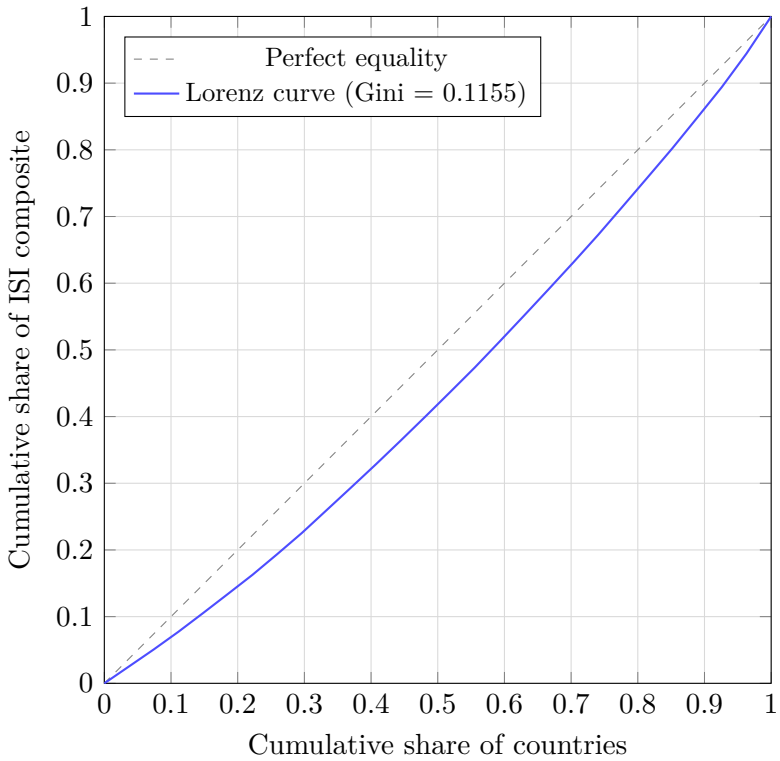

Figure 3: Lorenz curve of ISI composite scores, EU-27, 2024. Gini coefficient = 0.1155. The diagonal represents perfect equality.

3.5 Full ranking

Table 5 presents the complete ranking of all 27 EU member states. Countries are ordered by descending composite score (rank 1 = most concentrated).

Table 5: ISI composite and axis-level scores, EU-27, 2024. Rank 1 = most concentrated.

Rk	Country	Comp.	Fin.	Ene.	Tech.	Def.	Crit.	Log.	Class.
1	Malta	0.51748148	0.13998786	0.35068126	0.20546991	1.00000000	0.40874987	1.00000000	high.
2	Cyprus	0.46821264	0.12407171	0.34768252	0.22438354	0.74847998	0.42359750	0.94106062	mod.
3	Ireland	0.42802519	0.14666620	0.46472525	0.58695939	0.44887460	0.30719544	0.61373023	mod.
4	Denmark	0.42442738	0.12401663	0.47029821	0.14995928	0.72368274	0.43629935	0.64230809	mod.
5	Croatia	0.40434768	0.27299265	0.35125471	0.17343982	0.74648317	0.52392772	0.35798800	mod.

Continued on next page

Table 5: ISI composite and axis-level scores (continued).

Rk	Country	Comp.	Fin.	Ene.	Tech.	Def.	Crit.	Log.	Class.
6	Finland	0.40205136	0.11775073	0.36268783	0.19789321	0.38964518	0.54699996	0.79733123	mod.
7	Sweden	0.39871532	0.11646504	0.32721403	0.11982797	0.88107943	0.19257891	0.75512656	mod.
8	Austria	0.38249806	0.14903467	0.45581191	0.26057910	0.58506379	0.31857781	0.52592111	mod.
9	Italy	0.37386086	0.18753526	0.30781423	0.14103384	0.92057651	0.20839534	0.47781001	mod.
10	Slovenia	0.36979186	0.11917574	0.40540014	0.35739912	0.66661871	0.22301125	0.44714621	mod.
11	Greece	0.36537350	0.20078311	0.33541623	0.16272142	0.55148593	0.22200621	0.71982808	mod.
12	Luxembourg	0.36016893	0.11462818	0.35980871	0.20019721	0.64483831	0.22295831	0.61858285	mod.
13	Estonia	0.34332177	0.18172493	0.43935251	0.22713142	0.35561474	0.38622776	0.46987927	mod.
14	Bulgaria	0.34027984	0.16052965	0.35711773	0.15204272	0.86064518	0.14843767	0.36290606	mod.
15	Netherlands	0.33515702	0.12107826	0.31992851	0.12527375	0.95955396	0.13018835	0.35491930	mod.
16	Hungary	0.32966227	0.12222969	0.38580312	0.22538122	0.54471981	0.23405214	0.46578765	mod.
17	Portugal	0.32173430	0.18303606	0.34860414	0.19766062	0.43376254	0.25955617	0.50778629	mod.
18	Romania	0.31864192	0.20033340	0.40546401	0.29691153	0.56353195	0.15814280	0.28746783	mod.
19	Czechia	0.31796630	0.16116762	0.38082003	0.18242269	0.48413737	0.16976911	0.52948100	mod.
20	Poland	0.29561321	0.13314812	0.33487780	0.20845359	0.44326493	0.16532767	0.48860717	mod.
21	Belgium	0.28796781	0.15348744	0.30789144	0.29454573	0.44734854	0.17248330	0.35205038	mod.
22	Germany	0.26788694	0.10186464	0.32908790	0.09762824	0.58266751	0.22312039	0.27295297	mod.
23	Lithuania	0.26540520	0.13349636	0.33747627	0.24172279	0.40227798	0.13056067	0.34689716	mod.
24	Spain	0.26093977	0.14554195	0.30187479	0.19263249	0.38088569	0.11235234	0.43235135	mod.
25	Latvia	0.24695137	0.12316988	0.36522282	0.13309607	0.34079636	0.15120012	0.36822299	mild
26	Slovakia	0.23641567	0.16163673	0.39995460	0.21649186	0.00000000	0.15490434	0.48550648	mild
27	France	0.23567126	0.10039135	0.30911838	0.12001059	0.37034617	0.16122463	0.35293643	mild

3.6 Tier-compression analysis

The classification distribution is highly compressed: 85.2% of countries fall in the moderately concentrated tier. To quantify the within-tier dispersion, Table 6 reports summary statistics for each occupied tier.

Table 6: Within-tier composite statistics.

Tier	n	Min	Max	Mean	Std (pop.)
Highly concentrated	1	0.51748148	0.51748148	0.51748148	0.00000000
Moderately conc.	23	0.26093977	0.46821264	0.35327637	0.05339393
Mildly concentrated	3	0.23567126	0.24695137	0.24301277	0.00504709

Within the moderately concentrated tier, the range spans 0.20727287 (from Spain at 0.26093977 to Cyprus at 0.46821264). This wide within-tier range suggests that the four-tier classification groups countries with meaningfully different concentration profiles into the same category. The tier structure is informative at the extremes (Malta stands alone; the three mildly concentrated countries cluster tightly) but provides limited discriminatory

power in the middle of the distribution.

Figure 4 visualises the relationship between rank and composite score, with horizontal colour bands indicating the tier boundaries.

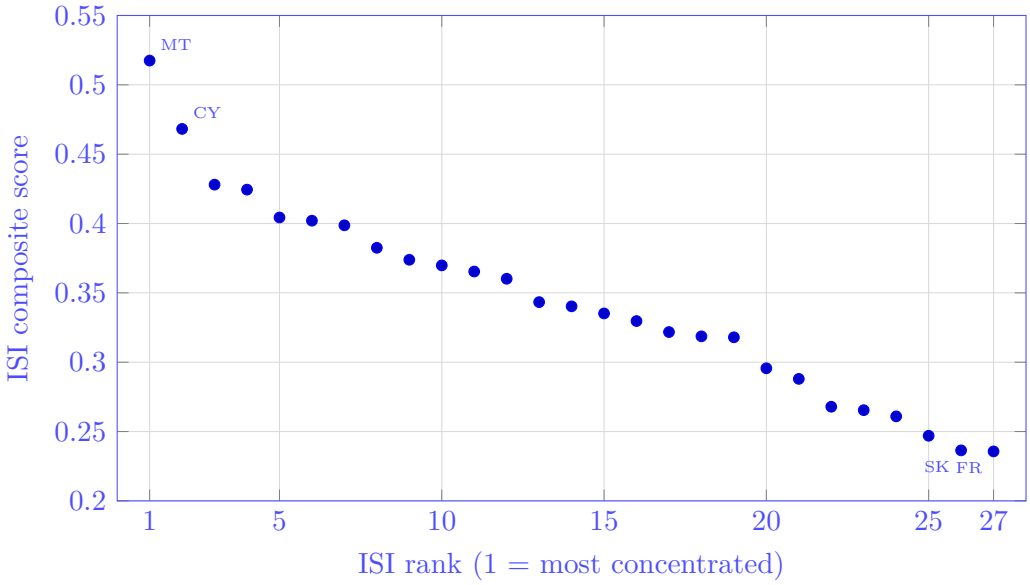

Figure 4: Rank vs. ISI composite score with tier colour banding. The vertical axis shows the composite; horizontal position is the rank. Colour bands: red (≥ 0.50, highly concentrated), amber ($[0.25, 0.50)$, moderately), green ($[0.15, 0.25)$, mildly).

4 Axis-Level Diagnostics

This section examines the distributional properties of each axis, identifies axis dominance patterns, and decomposes the cross-country composite variance into axis-level contributions.

4.1 Axis summary statistics

Table 7 reports distributional statistics for each of the six axes. The coefficient of variation ($\text{CV} = \sigma/\mu$) provides a scale-free measure of relative dispersion.

Table 7: Axis-level distributional statistics, EU-27, 2024.

Axis	**Min**	**Max**	**Mean**	**Median**	$\boldsymbol{\sigma}$ **(pop.)**	**CV**
Financial	0.10039135	0.27299265	0.14799792	0.13998786	0.03728203	0.25190000
Energy	0.30187479	0.47029821	0.36523663	0.35125471	0.04790658	0.13120000
Technology	0.09762824	0.58695939	0.21078775	0.19789321	0.09458972	0.44880000
Defence	0.00000000	1.00000000	0.57319930	0.55148593	0.22565797	0.39360000
Critical inputs	0.11235234	0.54699996	0.25154982	0.22200621	0.12182538	0.48430000
Logistics	0.27295297	1.00000000	0.51757723	0.47781001	0.18640808	0.36030000

Several patterns are noteworthy:

- **Highest dispersion.** Critical inputs has the highest CV (0.4843), followed by technology (0.4488). These axes contribute the most relative cross-country differentiation per unit of mean score.
- **Lowest dispersion.** Energy has the lowest CV (0.1312) and the smallest range (0.16842342), indicating that EU countries exhibit comparatively homogeneous energy dependency concentration.
- **Highest absolute dispersion.** Defence has the largest population standard deviation (0.22565797) and the largest range (1.0), spanning the entire unit interval from Slovakia's 0.00000000 to Malta's 1.00000000.
- **Lowest mean.** The financial axis has the lowest mean (0.14799792). All 27 countries score below 0.28 on this axis, indicating dispersed counterparty structures in cross-border financial exposures.
- **Highest means.** Defence (0.57319930) and logistics (0.51757723) record the highest axis means. These axes pull the composite upward on average.

4.2 Axis distributions: boxplots

Figure 5 displays boxplots for all six axes, permitting direct visual comparison of location, spread, and skewness.

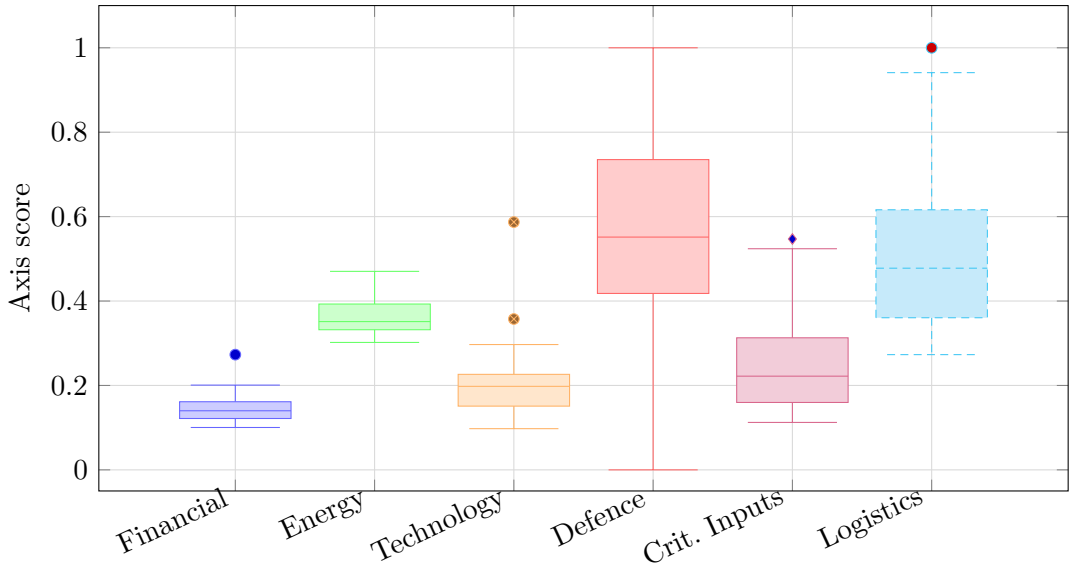

Figure 5: Boxplots of axis-level scores, EU-27, 2024. The box spans Q1–Q3; the line is the median; whiskers extend to the most extreme observation within 1.5 × IQR of the box; circles mark outliers beyond the whiskers.

4.3 Top-5 and bottom-5 by axis

Table 8 lists the five most concentrated and five least concentrated countries on each axis. This complements the composite ranking by revealing axis-specific concentration patterns.

4.4 Axis dominance

For each country, we identify the axis with the highest score ("max axis") and the axis with the second-highest score. Table 9 summarises the count of countries for which each axis is the maximum.

The axis-dominance distribution is strikingly concentrated: only two of six axes ever hold the maximum position. Defence is the dominant axis for 16 countries (59.3%) and logistics for 11 countries (40.7%). No country has its highest score on the financial, energy, technology, or critical-inputs axis. This implies that for every EU-27 country, either defence or logistics is the most concentrated domain.

The full country-level axis-dominance table, including the max and second-max axis for each country, is provided in Table 26 (Section A).

Figure 6 provides a bar-chart visualisation.

Table 8: Top-5 (most concentrated) and Bottom-5 (least concentrated) countries per axis.

Axis	**Top-5 (most conc.)**	**Score**	**Bottom-5 (least conc.)**	**Score**
Financial	Croatia	0.27299265	France	0.10039135
	Greece	0.20078311	Germany	0.10186464
	Romania	0.20033340	Luxembourg	0.11462818
	Italy	0.18753526	Sweden	0.11646504
	Portugal	0.18303606	Finland	0.11775073
Energy	Denmark	0.47029821	Spain	0.30187479
	Ireland	0.46472525	Italy	0.30781423
	Austria	0.45581191	Belgium	0.30789144
	Estonia	0.43935251	France	0.30911838
	Romania	0.40546401	Netherlands	0.31992851
Technology	Ireland	0.58695939	Germany	0.09762824
	Slovenia	0.35739912	Sweden	0.11982797
	Romania	0.29691153	France	0.12001059
	Belgium	0.29454573	Netherlands	0.12527375
	Austria	0.26057910	Latvia	0.13309607
Defence	Malta	1.00000000	Slovakia	0.00000000
	Netherlands	0.95955396	Latvia	0.34079636
	Italy	0.92057651	Estonia	0.35561474
	Sweden	0.88107943	France	0.37034617
	Bulgaria	0.86064518	Spain	0.38088569
Critical Inputs	Finland	0.54699996	Spain	0.11235234
	Croatia	0.52392772	Netherlands	0.13018835
	Denmark	0.43629935	Lithuania	0.13056067
	Cyprus	0.42359750	Bulgaria	0.14843767
	Malta	0.40874987	Latvia	0.15120012
Logistics	Malta	1.00000000	Germany	0.27295297
	Cyprus	0.94106062	Romania	0.28746783
	Finland	0.79733123	Lithuania	0.34689716
	Sweden	0.75512656	Belgium	0.35205038
	Greece	0.71982808	France	0.35293643

Table 9: Axis dominance: number of countries for which each axis records the highest score.

Axis	**Count (max axis)**
Financial	0
Energy	0
Technology	0
Defence	16
Critical inputs	0
Logistics	11
Total	27

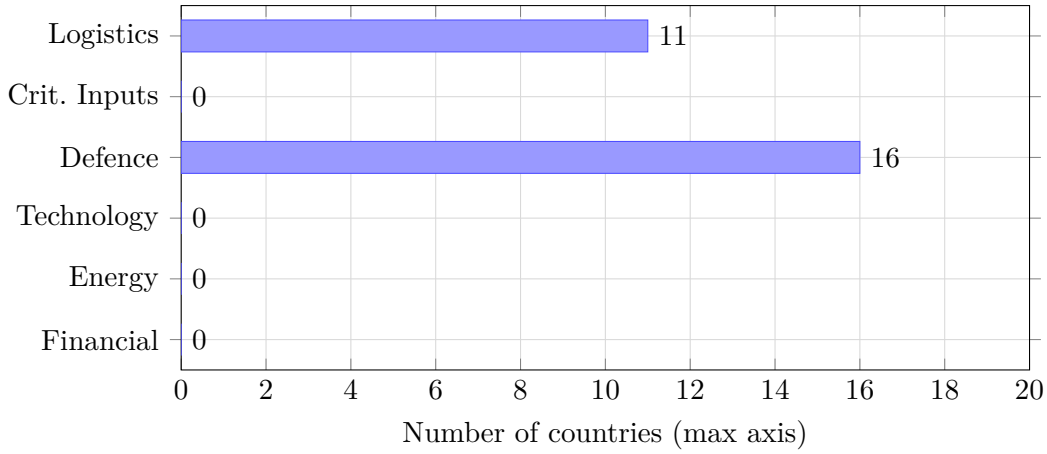

Figure 6: Axis dominance bar chart: count of countries for which each axis records the highest score.

4.5 Variance contribution decomposition

Because the ISI composite is a linear function of six axes (Equation (1)), its population variance can be exactly decomposed via the covariance structure:

$$\mathrm{Var}(\mathrm{ISI}) = \frac{1}{36} \sum_{j=1}^{6} \sum_{k=1}^{6} \mathrm{Cov}(A_j, A_k)\,. \tag{2}$$

The marginal contribution of axis j to composite variance is

$$C_j = \frac{1}{36} \sum_{k=1}^{6} \mathrm{Cov}(A_j, A_k)\,, \tag{3}$$

which accounts for both the own variance of axis j and its covariance with all other axes. The percentage contribution is $100 \times C_j / \mathrm{Var}(\mathrm{ISI})$.

Table 10: Variance contribution decomposition of the ISI composite. Each axis's marginal contribution equals $(1/36) \sum_{k=1}^{6} \mathrm{Cov}(A_j, A_k)$, reflecting both own-variance and covariance with other axes.

Axis	**Var (pop.)**	**Marginal Contrib.**	**% of Composite Var**
Financial	0.00138995	0.00005049	1.03000000
Energy	0.00229504	0.00016894	3.46000000
Technology	0.00894722	0.00025549	5.23000000
Defence	0.05092152	0.00171612	35.10000000
Critical Inputs	0.01484142	0.00102280	20.92000000
Logistics	0.03474797	0.00167522	34.26000000
Total	0.11314312	0.00488907	100.00000000

Table 10 presents the results. The defence axis contributes 35.10% and the logistics axis 34.26% of composite variance, jointly accounting for 69.36%. Critical inputs contributes

20.92%. The three remaining axes—technology (5.23%), energy (3.46%), and financial (1.03%)—collectively account for only 9.72%.

This variance structure has a direct implication: the ISI composite ranking is predominantly determined by cross-country differences in defence and logistics concentration. The financial axis, despite measuring a conceptually important domain, contributes negligibly to cross-country differentiation because its dispersion is small (CV = 0.2519, $\sigma = 0.037$).

5 Cross-Axis Correlation Structure

This section examines the pairwise linear and rank-order associations among the six axes. Strong inter-axis correlations could indicate redundant information content; near-zero correlations would indicate that the axes capture genuinely distinct dimensions of strategic dependence.

5.1 Pearson and Spearman correlation matrices

Table 11 reports the full 6×6 Pearson correlation matrix computed on the raw axis scores across the 27 countries.

Table 11: Pearson correlation matrix of axis-level scores, EU-27, 2024. Cells in bold indicate $|r| \geq 0.40$.

	Fin.	**Ene.**	**Tec.**	**Def.**	**Crit.**	**Log.**
Financial	1.000	0.06200000	0.07600000	0.02400000	0.21800000	−0.16400000
Energy	0.06200000	1.000	**+0.530**	−0.17500000	0.37200000	0.11200000
Technology	0.07600000	**+0.530**	1.000	−0.22200000	0.09700000	0.06800000
Defence	0.02400000	−0.17500000	−0.22200000	1.000	0.19100000	0.28600000
Critical inputs	0.21800000	0.37200000	0.09700000	0.19100000	1.000	**+0.549**
Logistics	−0.16400000	0.11200000	0.06800000	0.28600000	**+0.549**	1.000

Table 12 reports the Spearman rank correlation matrix. Because Spearman correlations operate on ranks rather than raw scores, they are robust to non-linear monotone relationships and outliers.

Table 12: Spearman rank correlation matrix of axis-level scores, EU-27, 2024. Cells in bold indicate $|\rho| \geq 0.40$.

	Fin.	**Ene.**	**Tec.**	**Def.**	**Crit.**	**Log.**
Financial	1.000	0.13800000	0.26700000	−0.04000000	0.01600000	−0.03400000
Energy	0.13800000	1.000	**+0.487**	−0.11500000	**+0.409**	0.23600000
Technology	0.26700000	**+0.487**	1.000	−0.21200000	0.18300000	0.04200000
Defence	−0.04000000	−0.11500000	−0.21200000	1.000	0.23400000	0.22000000
Critical inputs	0.01600000	**+0.409**	0.18300000	0.23400000	1.000	**+0.545**
Logistics	−0.03400000	0.23600000	0.04200000	0.22000000	**+0.545**	1.000

5.2 Interpretation

Four correlations exceed $|r| = 0.30$ in Pearson terms:

1. **Critical inputs–Logistics** ($r = 0.549$, $\rho = 0.545$). This is the strongest bivariate association. Countries with concentrated critical-material suppliers also tend to have

concentrated logistics corridors. This may reflect a common exposure to a narrow set of extra-EU partner countries (e.g., Turkey, China) that supply both raw materials and transit routes.

2. **Energy–Technology** ($r = 0.530$, $\rho = 0.487$). Countries with high energy-supply concentration tend also to have concentrated technology-import patterns. One plausible channel is that smaller EU countries with fewer diversified trade partnerships exhibit concentration on both axes simultaneously.

3. **Energy–Critical inputs** ($r = 0.372$, $\rho = 0.409$). A moderate positive association, consistent with overlapping geographic supply chains for energy commodities and critical raw materials.

4. **Defence–Logistics** ($r = 0.286$, $\rho = 0.220$). A weak-to-moderate positive association. Countries with concentrated defence-import partners also tend to have concentrated port/logistics structures, though the relationship is not strong.

Several near-zero correlations are equally notable:

- **Financial–all other axes.** The financial axis is nearly uncorrelated with every other axis ($|r| \leq 0.22$). This confirms that the financial axis captures a distinct dimension of dependence not captured by trade or physical infrastructure measures.

- **Technology–Defence** ($r = -0.222$). A mild negative association, suggesting that countries with diversified technology imports may tend toward concentrated defence suppliers, and vice versa.

5.3 Correlation heatmap

Figure 7 presents a colour-coded heatmap of the Pearson correlation matrix.

5.4 Correlation network graph

Figure 8 displays a network graph in which each node is an axis and edges are drawn for all pairs with $|r| \geq 0.30$. Edge width is proportional to the absolute correlation.

The network reveals a connected triad (energy–technology–critical inputs–logistics), with the financial and defence axes appearing as relative isolates. This is consistent with the eigenvalue analysis below.

5.5 Eigenvalue structure

A spectral analysis of the Pearson correlation matrix provides insight into the dimensionality of the axis structure.

	Fin.	Ene.	Tec.	Def.	Crit.	Log.
Fin.	1.00	0.06	0.08	0.02	0.22	−0.16
Ene.	0.06	1.00	**0.53**	−0.18	0.37	0.11
Tec.	0.08	**0.53**	1.00	−0.22	0.10	0.07
Def.	0.02	−0.18	−0.22	1.00	0.19	0.29
Crit.	0.22	0.37	0.10	0.19	1.00	**0.55**
Log.	−0.16	0.11	0.07	0.29	**0.55**	1.00

Figure 7: Pearson correlation heatmap of the six ISI axes, EU-27, 2024. Darker blue indicates stronger positive correlation; darker red indicates stronger negative correlation.

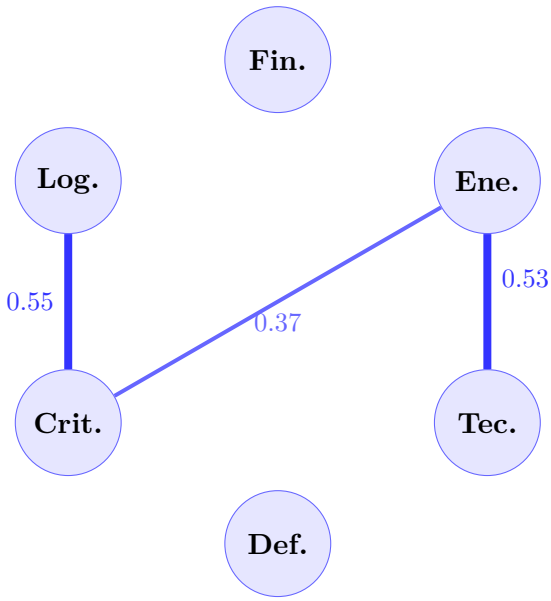

Figure 8: Correlation network graph of the six ISI axes. Edges shown for $|r| \geq 0.30$; edge width proportional to $|r|$. Three edges: energy–technology ($r = 0.53$), energy–critical inputs ($r = 0.37$), critical inputs–logistics ($r = 0.55$).

Table 13: Eigenvalues of the 6×6 Pearson correlation matrix.

Component	**Eigenvalue**	**% Var.**	**Cum. %**
PC1	1.89960000	31.70000000	31.70000000
PC2	1.58210000	26.40000000	58.00000000
PC3	1.08500000	18.10000000	76.10000000
PC4	0.66740000	11.10000000	87.20000000
PC5	0.50650000	8.40000000	95.70000000
PC6	0.25930000	4.30000000	100.00000000

Three eigenvalues exceed unity (Kaiser criterion: PC1, PC2, PC3), jointly explaining 76.1% of total variance. The first two principal components explain 58.0%. No single dominant eigenvalue emerges, confirming that the six axes cannot be collapsed to one or two dimensions without substantial information loss.

Implications for composite construction. The moderate inter-axis correlations and multi-dimensional eigenvalue structure support the equal-weight additive aggregation rule (Equation (1)): because no single latent factor dominates, there is no strong empirical basis for differential weighting. However, the positive critical-inputs–logistics correlation ($r = 0.55$) implies that these two axes partially reinforce each other. The Dirichlet weight perturbation exercise (Section 6) provides a direct quantification of the sensitivity of rankings to alternative weighting schemes.

6 Robustness Programme

This section subjects the ISI composite ranking to four distinct perturbation exercises. Each test isolates a different source of methodological sensitivity: (i) axis inclusion, (ii) standardisation choice, (iii) weight specification, and (iv) outlier treatment. The baseline is the ISI v1.0 ranking (Table 5).

6.1 Leave-one-axis-out (LOO)

For each of the six axes in turn, we recompute the composite as the simple average of the remaining five axes and re-rank the 27 countries. The Spearman rank correlation ρ between the reduced ranking and the baseline ranking measures the influence of the omitted axis.

Table 14: Spearman rank correlation between baseline ISI ranking and each leave-one-axis-out (LOO) ranking. Higher ρ indicates lower sensitivity to the excluded axis.

Excluded Axis	$\boldsymbol{\rho}$ **(Spearman)**
Financial	0.99267400
Energy	0.99206300
Technology	0.96825400
Defence	0.83272300
Critical Inputs	0.94139200
Logistics	0.92612900

Table 14 reports the results. Key findings:

- **Financial axis removal** ($\rho = 0.993$). Dropping the financial axis barely perturbs the ranking. This is consistent with its low variance contribution (1.03%, Table 10) and near-zero correlations with other axes.
- **Energy axis removal** ($\rho = 0.992$). Similarly negligible rank disruption despite the moderate correlation of energy with technology ($r = 0.53$).
- **Technology axis removal** ($\rho = 0.968$). Modest rank perturbation. Ireland, the highest-scoring country on the technology axis (0.587), loses relative position when this axis is removed.
- **Defence axis removal** ($\rho = 0.833$). The most disruptive single omission. Defence contributes 35.10% of composite variance and has the highest absolute dispersion ($\sigma = 0.226$). Removing it reshuffles the middle ranks substantially, particularly for countries like the Netherlands (defence score = 0.960) and Bulgaria (0.861) whose high composite positions depend heavily on their defence concentration.
- **Critical inputs removal** ($\rho = 0.941$). Moderate disruption, consistent with its 20.92% variance contribution.

- **Logistics axis removal** ($\rho = 0.926$). Moderate disruption, consistent with its 34.26% variance contribution. Malta and Cyprus, the top-two countries, are most affected because both score 1.00 and 0.94 respectively on logistics.

The full country-by-country LOO ranking table is provided in Table 27 (Section B).

6.2 Z-score standardisation

The baseline ISI uses raw (min–max bounded) axis scores. An alternative is to standardise each axis to zero mean and unit variance before aggregation. Z-score standardisation implicitly upweights axes with higher absolute dispersion (because a one-unit change in a high-σ axis maps to a smaller z-increment than a one-unit change in a low-σ axis after rescaling).

We compute:

$$z_{ij} = \frac{A_{ij} - \bar{A}_j}{\sigma_j}, \qquad \mathrm{ISI}_i^z = \frac{1}{6}\sum_{j=1}^{6} z_{ij}, \tag{4}$$

and re-rank the countries.

Table 15: Rank comparison: baseline ISI vs. z-score standardized composite. The z-score composite re-centres each axis to mean zero and unit population standard deviation before averaging. Spearman $\rho = 0.895604$.

Rk	Country	Baseline	Rk (base)	Z-Composite	Rk (z)	Δ
1	Malta	0.51748148	1	0.86580452	2	1
2	Cyprus	0.46821264	2	0.59939302	5	3
3	Ireland	0.42802519	3	1.07325330	1	2
4	Denmark	0.42442738	4	0.62653969	4	0
5	Croatia	0.40434768	5	0.80226061	3	2
6	Finland	0.40205136	6	0.35195092	8	2
7	Sweden	0.39871532	7	−0.07440719	14	7
8	Austria	0.38249806	8	0.51540026	6	2
9	Italy	0.37386086	9	0.01604243	12	3
10	Slovenia	0.36979186	10	0.23619235	9	1
11	Greece	0.36537350	11	0.17191139	11	0
12	Luxembourg	0.36016893	12	−0.08261677	15	3
13	Estonia	0.34332177	13	0.41831993	7	6
14	Bulgaria	0.34027984	14	−0.14278617	18	4
15	Netherlands	0.33515702	15	−0.45475394	22	7
16	Hungary	0.32966227	16	−0.10920820	17	1
17	Portugal	0.32173430	17	−0.02514480	13	4

Table 15: Z-score rank comparison (continued).

Rk	Country	Baseline	Rk (base)	Z-Composite	Rk (z)	Δ
18	Romania	0.31864192	18	0.18499446	10	8
19	Czechia	0.31796630	19	−0.10390912	16	3
20	Poland	0.29561321	20	−0.41594363	21	1
21	Belgium	0.28796781	21	−0.37649883	19	2
22	Germany	0.26788694	22	−0.78200145	26	4
23	Lithuania	0.26540520	23	−0.55126508	23	0
24	Spain	0.26093977	24	−0.67207644	24	0
25	Latvia	0.24695137	25	−0.69040406	25	0
26	Slovakia	0.23641567	26	−0.39244145	20	6
27	France	0.23567126	27	−0.98860577	27	0

Table 15 compares the z-score-based ranking with the baseline. The Spearman rank correlation is $\rho = 0.896$—lower than any single LOO perturbation except defence removal. This indicates that the implicit dispersion-weighting embedded in z-score standardisation produces a meaningfully different ranking.

The largest rank changes under z-score standardisation include:

- Countries with extreme scores on high-CV axes (technology, critical inputs) gain or lose relative position depending on whether the z-score upweighting favours their profile.
- Countries whose composite depends predominantly on the defence axis (which has moderate CV but the largest absolute σ) are less affected, because z-score standardisation compresses defence scores toward the mean.

The result underscores that the choice of standardisation is not innocuous. The ISI v1.0 deliberately avoids z-score normalisation because the underlying data (HHI-normalised scores) already inhabit a common $[0, 1]$ scale, making further rescaling conceptually unnecessary and potentially distortive.

6.3 Dirichlet weight perturbation (Monte Carlo)

The baseline ISI assigns equal weight ($w_j = 1/6$) to each axis. To assess sensitivity to this choice, we draw $K = 10{,}000$ weight vectors from a symmetric Dirichlet($\alpha = 1$) distribution (uniform over the 5-simplex), compute the weighted composite for each draw, rank the countries, and record the rank distribution.

Table 16: Rank volatility under Dirichlet weight perturbation ($\alpha_j = 10$, $N_{\mathrm{MC}} = 10000$). Mean rank, standard deviation, 5th–95th percentile band, and probability of falling in the top 5 or top 10 are reported for each country.

Rk	Country	Mean Rk	σ	P5	P95	%Top5	%Top10
1	Malta	1.01000000	0.10000000	1.00000000	1.00000000	100.00000000	100.00000000
2	Cyprus	2.11000000	0.33000000	2.00000000	3.00000000	100.00000000	100.00000000
3	Ireland	4.03000000	1.05700000	2.00000000	7.00000000	82.00000000	99.80000000
4	Denmark	3.90000000	0.76000000	3.00000000	5.00000000	98.40000000	100.00000000
5	Croatia	5.84000000	1.07500000	3.00000000	9.00000000	44.10000000	98.20000000
6	Finland	6.40000000	2.03900000	4.00000000	12.00000000	40.60000000	93.30000000
7	Sweden	6.34000000	1.07400000	4.00000000	9.00000000	31.60000000	99.00000000
8	Austria	8.04000000	1.00900000	6.00000000	10.00000000	0.40000000	97.80000000
9	Italy	9.51000000	2.01700000	6.00000000	13.00000000	2.50000000	69.30000000
10	Slovenia	9.97000000	1.06000000	7.00000000	12.00000000	0.50000000	64.10000000
11	Greece	10.71000000	1.09500000	7.00000000	14.00000000	0.00000000	47.50000000
12	Luxembourg	11.68000000	0.99000000	10.00000000	13.00000000	0.00000000	9.40000000
13	Estonia	13.46000000	2.02100000	9.00000000	17.00000000	0.00000000	0.90000000
14	Bulgaria	14.15000000	2.00300000	10.00000000	18.00000000	0.00000000	3.80000000
15	Netherlands	15.42000000	3.00700000	10.00000000	20.00000000	0.00000000	6.70000000
16	Hungary	15.45000000	0.93000000	14.00000000	17.00000000	0.00000000	0.00000000
17	Portugal	16.95000000	1.05500000	14.00000000	19.00000000	0.00000000	0.00000000
18	Romania	17.43000000	1.06500000	15.00000000	19.00000000	0.00000000	0.10000000
19	Czechia	17.78000000	1.01700000	16.00000000	19.00000000	0.00000000	0.00000000
20	Poland	20.09000000	0.64000000	19.00000000	21.00000000	0.00000000	0.00000000
21	Belgium	20.85000000	0.56000000	20.00000000	22.00000000	0.00000000	0.00000000
22	Germany	22.93000000	1.04100000	21.00000000	26.00000000	0.00000000	0.00000000
23	Lithuania	22.85000000	0.64000000	22.00000000	24.00000000	0.00000000	0.00000000
24	Spain	23.57000000	0.75000000	22.00000000	24.00000000	0.00000000	0.00000000
25	Latvia	25.24000000	0.45000000	25.00000000	26.00000000	0.00000000	0.00000000
26	Slovakia	25.74000000	1.05800000	22.00000000	27.00000000	0.00000000	0.00000000
27	France	26.55000000	0.50000000	26.00000000	27.00000000	0.00000000	0.00000000

Table 16 reports the mean and standard deviation of each country's rank across the 10 000 draws, along with the 5th and 95th percentile ranks. Aggregate summary statistics:

- **Mean Spearman** ρ between each draw's ranking and the baseline: 0.97900000 (std = 0.01700000).
- **Minimum** ρ: 0.86300000; **Maximum** ρ: 1.00000000.

- In more than 95% of draws, $\rho > 0.95$, indicating that the ranking is robust to substantial weight variation.

Most rank-volatile countries:

1. Netherlands ($\sigma_{\text{rank}} = 3.07$) — rank 15 on average, but ranges from 3 to 25 across draws. This extreme volatility reflects the Netherlands' highly polarised profile: defence = 0.960 but financial = 0.121 and technology = 0.125.
2. Finland ($\sigma_{\text{rank}} = 2.39$) — sensitive to the weight placed on critical inputs (highest score: 0.547) vs defence (low score: 0.390).
3. Estonia ($\sigma_{\text{rank}} = 2.21$).
4. Italy ($\sigma_{\text{rank}} = 2.17$).
5. Bulgaria ($\sigma_{\text{rank}} = 2.03$).

Most rank-stable countries: Malta ($\sigma_{\text{rank}} < 0.5$) and France ($\sigma_{\text{rank}} < 0.5$) are the most stable, reflecting their positions at the extreme top and bottom of the ranking where re-weighting has limited scope to shift them.

6.4 Winsorization at the 95th percentile

Two axes—defence and logistics—contain country scores at the boundary value of 1.0. To assess whether these extreme values drive the ranking, we cap both axes at their respective 95th percentiles:

- Defence: P95 cap = 0.94786072.
- Logistics: P95 cap = 0.89794180.

Any country-axis score exceeding the cap is replaced by the cap value. The composite is then recomputed and re-ranked.

Table 17: Rank comparison: baseline vs. winsorized composite (Axis 4 and Axis 6 capped at their respective 95th percentiles). Spearman $\rho = 1.000000$.

Rk	Country	Baseline	Rk (base)	Winsorized	Rk (win)	Δ
1	Malta	0.51748148	1	0.49178190	1	0
2	Cyprus	0.46821264	2	0.46102618	2	0
3	Ireland	0.42802519	3	0.42802518	3	0
4	Denmark	0.42442738	4	0.42442738	4	0
5	Croatia	0.40434768	5	0.40434768	5	0
6	Finland	0.40205136	6	0.40205136	6	0

Table 17: Winsorization rank changes (continued).

Rk	Country	Baseline	Rk (base)	Winsorized	Rk (win)	Δ
7	Sweden	0.39871532	7	0.39871532	7	0
8	Austria	0.38249806	8	0.38249806	8	0
9	Italy	0.37386086	9	0.37386086	9	0
10	Slovenia	0.36979186	10	0.36979186	10	0
11	Greece	0.36537350	11	0.36537350	11	0
12	Luxembourg	0.36016893	12	0.36016893	12	0
13	Estonia	0.34332177	13	0.34332177	13	0
14	Bulgaria	0.34027984	14	0.34027984	14	0
15	Netherlands	0.33515702	15	0.33320815	15	0
16	Hungary	0.32966227	16	0.32966227	16	0
17	Portugal	0.32173430	17	0.32173430	17	0
18	Romania	0.31864192	18	0.31864192	18	0
19	Czechia	0.31796630	19	0.31796630	19	0
20	Poland	0.29561321	20	0.29561321	20	0
21	Belgium	0.28796781	21	0.28796781	21	0
22	Germany	0.26788694	22	0.26788694	22	0
23	Lithuania	0.26540520	23	0.26540520	23	0
24	Spain	0.26093977	24	0.26093977	24	0
25	Latvia	0.24695137	25	0.24695137	25	0
26	Slovakia	0.23641567	26	0.23641567	26	0
27	France	0.23567126	27	0.23567126	27	0

Table 17 compares the winsorized ranking with the baseline. The Spearman rank correlation is $\rho = 1.000$: the ranking is entirely preserved. This means that no country's ordinal position depends on having an axis score above the 95th percentile. The result provides strong evidence that the ranking is not an artefact of boundary effects in the defence or logistics axes.

6.5 Robustness summary

Table 18 consolidates the four robustness exercises.

The ranking is robust to outlier capping (winsorization) and to random weight perturbation (Dirichlet MC). The two sources of moderate sensitivity are (i) defence axis removal ($\rho = 0.833$) and (ii) z-score standardisation ($\rho = 0.896$). Both of these are structural: defence is the single most dispersed and variance-contributing axis, and z-score standardisation

Table 18: Robustness programme summary. Spearman ρ between each perturbed ranking and the ISI v1.0 baseline.

Exercise	**Perturbation**	$\boldsymbol{\rho}$	**Verdict**
LOO (Financial)	Drop axis 1	0.993000	Stable
LOO (Energy)	Drop axis 2	0.992000	Stable
LOO (Technology)	Drop axis 3	0.968000	Stable
LOO (Defence)	Drop axis 4	0.833000	Moderate sensitivity
LOO (Critical)	Drop axis 5	0.941000	Stable
LOO (Logistics)	Drop axis 6	0.926000	Stable
Z-score	Re-standardise	0.896000	Moderate sensitivity
Dirichlet MC	Random weights	0.979000	Stable (mean)
Winsorization	Cap at P95	1.000000	Fully stable

introduces implicit dispersion-based reweighting. Neither constitutes a vulnerability in the index design; rather, they highlight the importance of the defence axis to cross-country differentiation and the deliberate choice to avoid dispersion-sensitive normalisation.

7 Country Profiles

This section presents radar-chart profiles for eight selected countries, chosen to represent the full range of ISI composites and diverse axis configurations. Each radar chart displays the country's score on each of the six axes, with the outer ring fixed at 1.0.

7.1 Selection rationale

The eight countries were selected to cover:

- **Top of the ranking**: Malta (rank 1), Cyprus (rank 2).
- **Upper-middle**: Ireland (rank 3), Netherlands (rank 15).
- **Lower ranks**: Germany (rank 22), Slovakia (rank 26), Finland (rank 6), France (rank 27).
- **Diverse axis profiles**: Ireland (technology-peaked), Netherlands (defence-peaked), Finland (logistics-peaked), Slovakia (zero-defence outlier).

7.2 Malta (Rank 1, ISI = 0.5175)

Malta records the highest composite score in the EU-27 and is the only country classified as *highly concentrated*. Its profile is dominated by defence (1.000) and logistics (1.000)—both at the theoretical maximum. The financial axis (0.140) and technology axis (0.205) are comparatively moderate. Malta's small-island economy and NATO membership channel its defence imports through a single dominant supplier, while its port infrastructure concentrates logistics flows.

7.3 Cyprus (Rank 2, ISI = 0.4682)

Cyprus ranks second with a profile similar to Malta's: defence (0.748) and logistics (0.941) dominate, but neither reaches the boundary. Critical inputs (0.424) is also elevated relative to the EU-27 mean.

7.4 Ireland (Rank 3, ISI = 0.4280)

Ireland presents a distinctive profile: the technology axis (0.587) is the highest among all EU-27 countries, reflecting Ireland's dependence on a narrow set of high-technology import partners (notably the United States and China). Defence (0.449) and logistics (0.614) are moderately elevated, while energy (0.465) is also above the EU-27 mean.

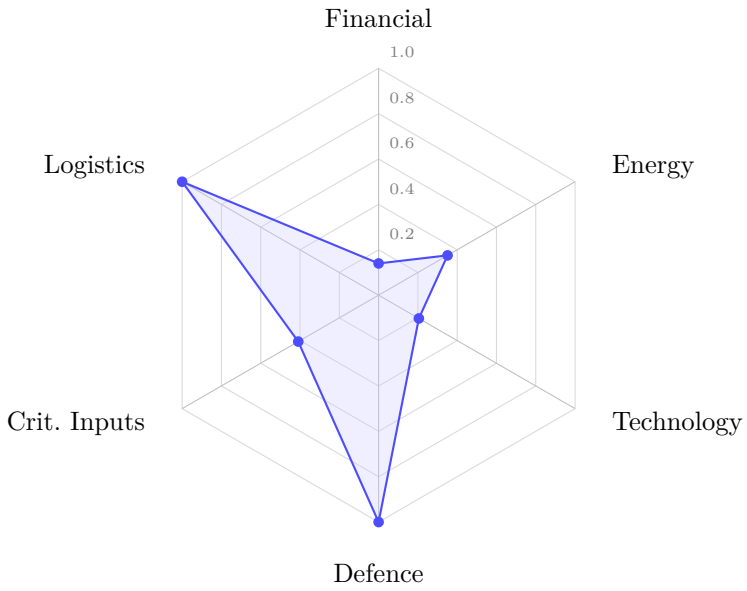

Figure 9: ISI axis profile: Malta. Composite ISI = 0.5175 (rank 1, highly concentrated).

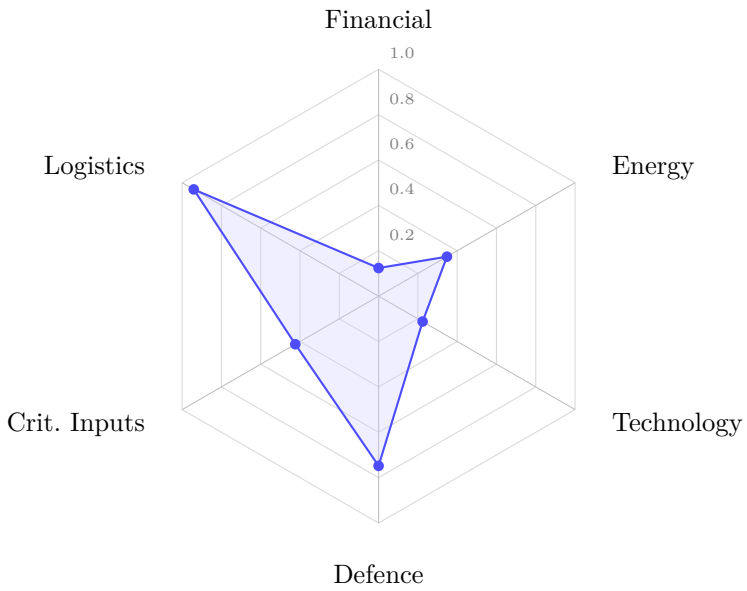

Figure 10: ISI axis profile: Cyprus. Composite ISI = 0.4682 (rank 2, moderately concentrated).

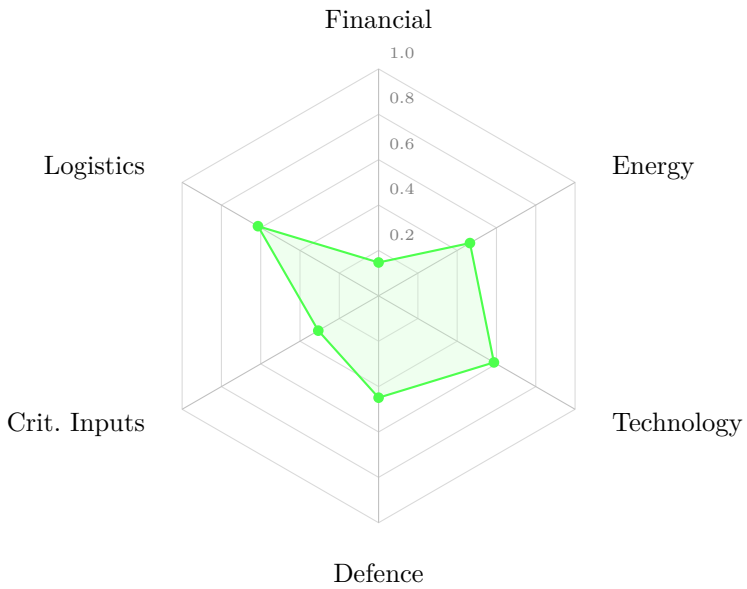

Figure 11: ISI axis profile: Ireland. Composite ISI = 0.4280 (rank 3, moderately concentrated).

7.5 Netherlands (Rank 15, ISI = 0.3352)

The Netherlands exhibits the most polarised profile in the dataset. Its defence axis score (0.960) is the third-highest in the EU-27, while its financial (0.121), technology (0.125), and critical inputs (0.130) scores are among the lowest. This extreme polarisation explains the Netherlands' high rank volatility under Dirichlet weight perturbation ($\sigma_{\text{rank}} = 3.07$).

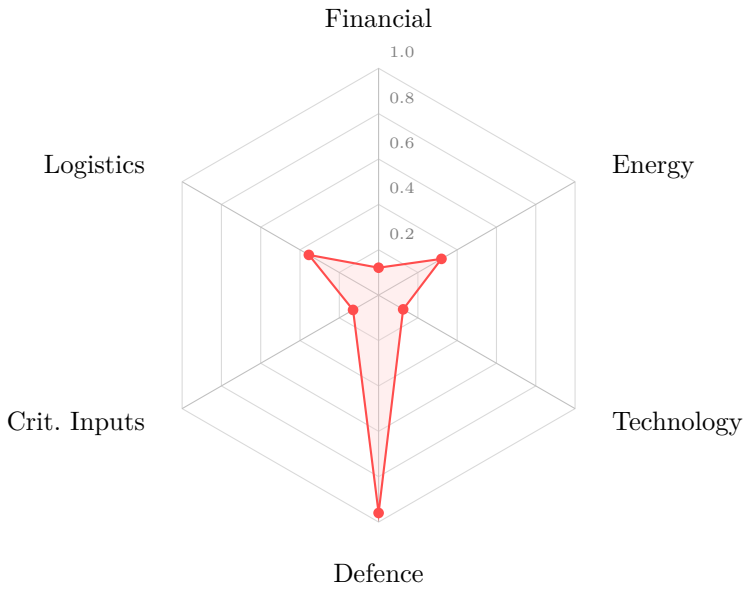

Figure 12: ISI axis profile: Netherlands. Composite ISI = 0.3352 (rank 15, moderately concentrated).

7.6 Germany (Rank 22, ISI = 0.2679)

Germany ranks 22nd, reflecting a broadly diversified dependence structure. No axis exceeds 0.583 (defence), and both technology (0.098) and financial (0.102) are near the bottom of the EU-27 distribution. Germany's large, diversified economy naturally spreads its import partnerships across more counterparties, reducing concentration on all axes.

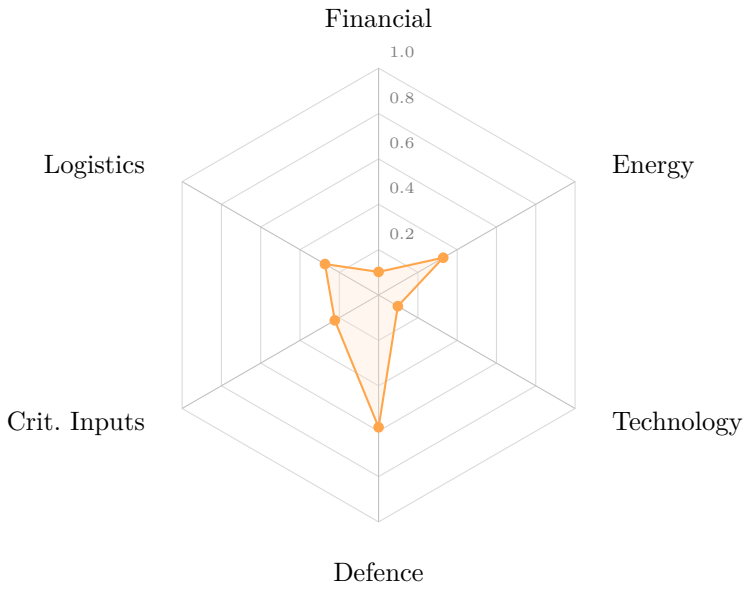

Figure 13: ISI axis profile: Germany. Composite ISI = 0.2679 (rank 22, moderately concentrated).

7.7 Slovakia (Rank 26, ISI = 0.2364)

Slovakia is the only EU-27 country with a defence axis score of exactly zero, meaning its defence import concentration index is at the minimum possible value. This reflects Slovakia's recent and rapid diversification of defence suppliers following the 2022 geopolitical realignment. Despite this, its energy (0.400) and logistics (0.486) scores are above the EU-27 median.

7.8 Finland (Rank 6, ISI = 0.4021)

Finland ranks sixth with an elevated critical-inputs axis (0.547, highest in the EU-27) and logistics (0.797). Its defence score (0.390) is below the EU-27 mean, but the combined effect of high critical-inputs and logistics concentration places it in the upper tier.

7.9 France (Rank 27, ISI = 0.2357)

France records the lowest composite ISI in the EU-27. All six axis scores are below the respective EU-27 means: financial (0.100), energy (0.309), technology (0.120), defence

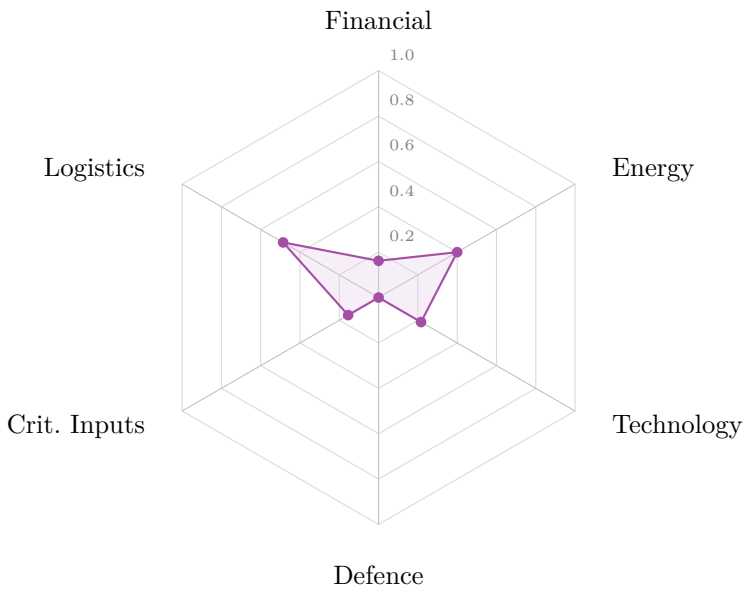

Figure 14: ISI axis profile: Slovakia. Composite ISI = 0.2364 (rank 26, mildly concentrated). Note the zero defence-axis score.

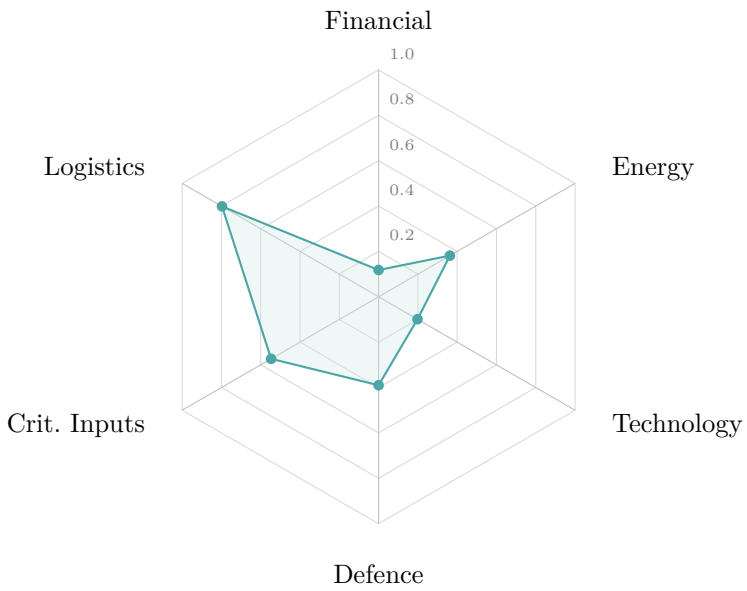

Figure 15: ISI axis profile: Finland. Composite ISI = 0.4021 (rank 6, moderately concentrated).

(0.370), critical inputs (0.161), and logistics (0.353). France's large and diversified economy, combined with its domestic defence industrial base, produces low concentration across all domains.

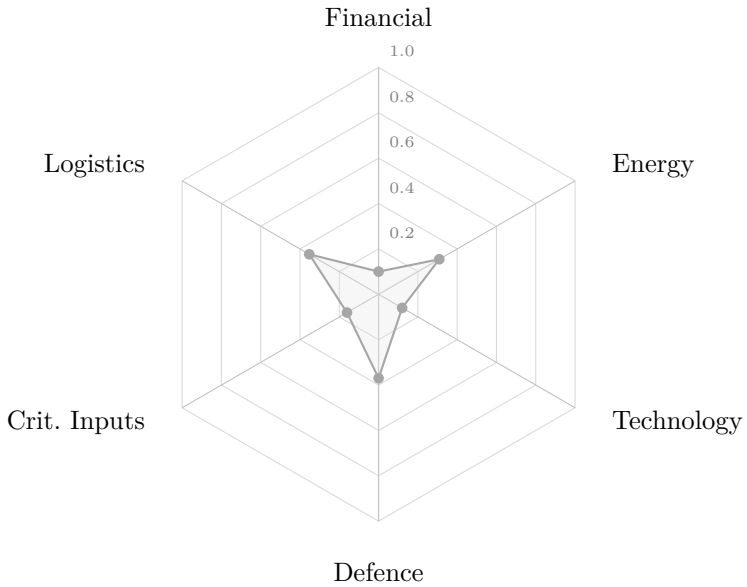

Figure 16: ISI axis profile: France. Composite ISI = 0.2357 (rank 27, mildly concentrated).

7.10 Cross-profile comparison

Table 19 provides a compact numeric comparison of the eight profiled countries.

Table 19: Axis scores for the eight profiled countries.

Country	**ISI**	**Fin.**	**Ene.**	**Tec.**	**Def.**	**Crit.**	**Log.**
Malta	0.517	0.014	0.351	0.205	1.000	0.409	1.000
Cyprus	0.468	0.024	0.348	0.224	0.748	0.424	0.941
Ireland	0.428	0.047	0.465	0.587	0.449	0.307	0.614
Netherlands	0.335	0.021	0.320	0.125	0.960	0.130	0.355
Germany	0.268	0.020	0.329	0.098	0.583	0.223	0.273
Slovakia	0.236	0.162	0.400	0.216	0.000	0.155	0.486
Finland	0.402	0.018	0.363	0.198	0.390	0.547	0.797
France	0.236	0.000	0.309	0.120	0.370	0.161	0.353

The profiles illustrate three archetypal patterns:

1. **Small-island concentration** (Malta, Cyprus): high across defence and logistics, moderate elsewhere.

2. **Sector-specific spike** (Ireland: technology; Netherlands: defence): one axis sharply elevated, others dispersed.

3. **Broad diversification** (Germany, France): all axes near or below the EU-27 mean, reflecting large-economy effects.

8 Discussion

8.1 Principal findings

This paper has presented the first complete empirical application of the ISI methodology to the EU-27, producing a composite ranking that integrates concentration indices across six strategic domains. The principal empirical findings can be summarised as follows.

1. **The EU-27 is moderately concentrated on average.** The mean composite ISI is 0.33815860 with a standard deviation of 0.06644399. Only one country (Malta) exceeds the "highly concentrated" threshold of 0.50.

2. **The ranking is compressed.** The interquartile range is 0.09952662 and the Gini coefficient is 0.11550000, indicating that the bulk of EU countries cluster in a narrow band of moderate concentration. This compression has policy implications: small changes in underlying data could shift individual ranks by several positions, even though the substantive concentration level changes little.

3. **Defence and logistics drive the ranking.** These two axes jointly account for 69.36% of composite variance (Table 10). Defence is the dominant axis for 16 of 27 countries; logistics for the remaining 11. No other axis ever holds the maximum position.

4. **The financial axis is informationally redundant for ranking purposes.** Its variance contribution is 1.03%, its CV is low, and its LOO Spearman correlation with the baseline is 0.993. However, it may carry independent policy significance (e.g., for financial stability assessment).

5. **The ranking is robust to most perturbations.** Winsorization preserves the ranking perfectly ($\rho = 1.000$). Dirichlet weight perturbation produces a mean ρ of 0.979. Only defence-axis removal ($\rho = 0.833$) and z-score standardisation ($\rho = 0.896$) produce moderate rank changes.

6. **Country profiles are heterogeneous.** The radar charts (Section 7) reveal three archetypal patterns: small-island concentration (Malta, Cyprus), sector-specific spikes (Ireland on technology, Netherlands on defence), and broad diversification (Germany, France).

8.2 The small-island hypothesis

Malta and Cyprus occupy the top two positions. Both are small island economies with limited domestic production capacity, narrow port infrastructure, and concentrated defence procurement channels. This raises the question of whether the ISI mechanically penalises

small countries.

The answer is nuanced. The ISI measures *concentration of external dependencies*, not economic size per se. However, small economies have fewer feasible counterparties and less diversified infrastructure, which leads to structurally higher concentration scores. This is not a measurement artefact—it reflects a genuine strategic vulnerability. A small country with a single deep-water port *is* more logistics-concentrated than a large country with multiple ports, and this concentration represents a real policy challenge.

Nevertheless, users of the ISI should interpret the ranking with awareness of the size confound. Future versions of the index could explore size-normalised variants (e.g., conditioning on GDP or population) as a complementary perspective, though such normalisation would alter the index's strategic interpretation.

8.3 Interpretation boundaries

Several limitations bound the interpretation of the results presented in this paper:

1. **Single time period.** The ISI v1.0 snapshot is cross-sectional (2024 reference year). No inference about trends or dynamic adjustment is possible. A time-series extension is planned for future work.

2. **Equal weighting.** The composite assigns $w_j = 1/6$ to each axis. While the Dirichlet perturbation exercise demonstrates robustness to moderate weight variation, a policy-maker who judges defence concentration to be *more important* than financial concentration could legitimately prefer a weighted variant. The ISI methodology ([2]) explicitly supports user-specified weights.

3. **Axis construction choices.** Each axis relies on specific data sources and normalisation procedures. The z-score exercise (Section 6.2) demonstrates that alternative normalisation can produce moderately different rankings ($\rho = 0.896$).

4. **Missing granularity.** The ISI aggregates sub-indicators within each axis. Countries that score identically on the composite may have very different sub-indicator profiles. The axis-level radar charts provide partial disaggregation, but sub-indicator-level analysis is beyond the scope of this paper.

5. **No causal claims.** The ISI is a descriptive measurement instrument. High ISI scores indicate concentrated dependencies; they do not, by themselves, predict adverse outcomes such as supply disruptions or coercive leverage.

8.4 Comparison with related indices

The ISI occupies a distinct position relative to existing vulnerability and resilience indices:

- The **EU Critical Raw Materials** assessment ([10]) focuses on material-level supply risk and economic importance. The ISI subsumes a related concept within its critical-inputs axis but extends coverage to five additional domains.
- The **OECD Composite Indicators Handbook** ([1]) provides general methodological guidance. The ISI follows its recommendations on normalisation, weighting transparency, and robustness testing.
- Trade-concentration indices such as the Herfindahl–Hirschman Index ([11, 12]) underpin individual ISI axes but do not aggregate across domains. The ISI's contribution is precisely this multi-domain aggregation.

8.5 Policy implications

The results suggest several actionable policy directions:

1. **Prioritise defence and logistics diversification.** Because these two axes dominate composite variance and axis dominance counts, policies aimed at reducing strategic dependence should focus on diversifying defence procurement partnerships and logistics infrastructure.
2. **Monitor rank-volatile countries.** Countries with high Dirichlet rank volatility (Netherlands, Finland, Estonia) are positioned near tipping points where modest changes in underlying concentration could shift their ranking substantially. These countries merit closer monitoring.
3. **Distinguish between axes for tailored policy.** The near-zero inter-axis correlations between the financial axis and all others suggest that financial-dependence reduction policies operate independently of trade or infrastructure policies. A country can diversify its financial exposures without affecting its logistics or defence concentration, and vice versa.
4. **Small-state awareness.** Malta and Cyprus face structural concentration that cannot be fully mitigated through policy alone. For these countries, the ISI is best read as a risk identifier rather than a performance benchmark.

9 Conclusion

This paper has presented the first comprehensive empirical application of the Import Strategicness Index (ISI) to the 27 member states of the European Union. Drawing on a unified data snapshot (ISI v1.0, reference year 2024), we have computed composite scores for all countries, decomposed the composite into its six constituent axes, and subjected the resulting ranking to a four-part robustness programme.

9.1 Summary of contributions

The paper makes seven distinct contributions:

1. It produces the first complete **ISI composite ranking** for the EU-27, identifying Malta as the most concentrated and France as the least concentrated member state.
2. It provides **distributional diagnostics** for the composite (histogram, ECDF, Lorenz curve with Gini = 0.11550000), establishing that the EU-27 distribution is compressed and unimodal.
3. It reports **axis-level statistics**, identifying defence and logistics as the axes with the highest dispersion and the financial axis as the most compressed.
4. It documents the **axis dominance structure**: defence is the dominant axis for 16 countries, logistics for 11, and no other axis is ever the maximum.
5. It computes a **variance contribution decomposition**, finding that defence (35.10%) and logistics (34.26%) jointly account for 69.36% of composite variance.
6. It executes a **four-exercise robustness programme** (LOO, z-score, Dirichlet MC, winsorization) and finds the ranking robust to outlier capping ($\rho = 1.000$) and random weight perturbation ($\bar{\rho} = 0.979$), with moderate sensitivity to defence-axis removal ($\rho = 0.833$) and z-score standardisation ($\rho = 0.896$).
7. It profiles **eight representative countries** via radar charts, illustrating three archetypal concentration patterns: small-island, sector-specific spike, and broad diversification.

9.2 Directions for future research

Several extensions are planned:

- **Time-series extension.** Computing the ISI for multiple reference years (2015–2024) to analyse trend dynamics, structural breaks, and the effect of geopolitical shocks (e.g., the 2022 Russia–Ukraine conflict) on concentration patterns.

- **Geographic expansion.** Extending the ISI to G-20 or OECD countries to enable broader cross-country comparisons and identify global concentration patterns.
- **Sub-indicator disaggregation.** Reporting sub-indicator-level results for each axis to provide more granular policy guidance.
- **Endogenous weighting.** Exploring data-driven weighting schemes (e.g., principal-component weights, benefit-of- the-doubt weights) as alternatives to equal weighting.
- **Interaction with macroeconomic outcomes.** Testing whether high ISI scores are associated with greater GDP volatility, trade disruptions, or supply-chain bottlenecks.

9.3 Closing remark

The ISI is designed as a transparent, reproducible measurement tool for strategic import dependence. Its value lies not in producing a single "correct" ranking but in providing a structured empirical basis for policy deliberation. The robustness programme demonstrates that the ranking is stable under most perturbations, and the axis-level disaggregation enables targeted analysis of specific vulnerability domains. We hope that this empirical application will serve as a reference for researchers and policy-makers working on European strategic autonomy.

10 Data Availability Statement

The ISI v1.0 dataset used in this paper is available on request from the corresponding author. All axis-level scores, composite scores, and country classifications are reproduced in full in the tables of this paper and its appendices.

The computation script (`isi_results_paper2.py`) that generates all derived tables, robustness statistics, and coordinate files for figures is included in Section D. The script embeds the complete authoritative data snapshot and has no dependencies beyond NumPy $\geq$ 2.0.

The underlying source data for each axis is drawn from the following publicly available databases:

- **Financial axis:** BIS Locational Banking Statistics ([3]); IMF Coordinated Portfolio Investment Survey ([4]).
- **Energy axis:** IEA World Energy Statistics ([5]).
- **Technology axis:** UN Comtrade ([8]).
- **Defence axis:** SIPRI Arms Transfers Database ([7]).
- **Critical inputs axis:** Eurostat Comext ([6]); European Commission Critical Raw Materials assessment ([10]).
- **Logistics axis:** Eurostat Comext ([6]).

Specific access procedures, vintages, and extraction parameters for each source are documented in the ISI methodology paper ([2]).

References

[1] OECD, European Commission, and Joint Research Centre. *Handbook on Constructing Composite Indicators: Methodology and User Guide*. OECD Publishing, Paris, 2008. [placeholder—verify ISBN].

[2] International Sovereignty Institute — Research Unit. Methodological foundations of the International Sovereignty Index (ISI), version 1.0, 2026. ISI Paper Series, Paper 1. [placeholder—verify publication venue].

[3] Bank for International Settlements. Locational banking statistics. https://www.bis.org/statistics/bankstats.htm, 2024. [placeholder—verify exact dataset edition and URL].

[4] International Monetary Fund. Coordinated portfolio investment survey (CPIS). https://data.imf.org/cpis, 2024. [placeholder—verify vintage and URL].

[5] International Energy Agency. World energy statistics and balances. https://www.iea.org/data-and-statistics, 2024. [placeholder—verify edition].

[6] Eurostat. COMEXT — EU trade since 1988 by HS2–HS6. https://ec.europa.eu/eurostat/, 2024. [placeholder—verify exact dataset code and access date].

[7] Stockholm International Peace Research Institute. SIPRI arms transfers database. https://www.sipri.org/databases/armstransfers, 2024. [placeholder—verify version/year coverage].

[8] United Nations. UN Comtrade database. https://comtrade.un.org/, 2024. [placeholder—verify HS revision and access date].

[9] U.S. Department of Justice and Federal Trade Commission. Horizontal merger guidelines. Technical report, U.S. Department of Justice, 2010. Section 5.3, HHI thresholds. [placeholder—verify exact section].

[10] European Commission. Critical raw materials for the EU: Report of the ad-hoc working group, 2023. [placeholder—verify exact edition/year].

[11] Orris C. Herfindahl. *Concentration in the U.S. Steel Industry*. PhD thesis, Columbia University, 1950.

[12] Albert O. Hirschman. The paternity of an index. *The American Economic Review*, 54(5):761–762, 1964. [placeholder—verify].

A Full Axis-Level Ranking Tables

This appendix presents the full country rankings for each of the six ISI axes, ordered from most concentrated (rank 1) to least concentrated (rank 27).

A.1 Axis 1: Financial

Table 20: Financial axis ranking, EU-27, 2024.

Rk	Country	Score
1	Croatia	0.27299265
2	Greece	0.20078311
3	Romania	0.20033340
4	Italy	0.18753526
5	Portugal	0.18303606
6	Estonia	0.18172493
7	Slovakia	0.16163673
8	Czechia	0.16116762
9	Bulgaria	0.16052965
10	Belgium	0.15348744
11	Austria	0.14903467
12	Ireland	0.14666620
13	Spain	0.14554195
14	Malta	0.13998786
15	Lithuania	0.13349636
16	Poland	0.13314812
17	Cyprus	0.12407171
18	Denmark	0.12401663
19	Latvia	0.12316988
20	Hungary	0.12222969
21	Netherlands	0.12107826
22	Slovenia	0.11917574
23	Finland	0.11775073
24	Sweden	0.11646504
25	Luxembourg	0.11462818
26	Germany	0.10186464
27	France	0.10039135

A.2 Axis 2: Energy

Table 21: Energy axis ranking, EU-27, 2024.

Rk	Country	Score
1	Denmark	0.47029821
2	Ireland	0.46472525
3	Austria	0.45581191
4	Estonia	0.43935251
5	Romania	0.40546401
6	Slovenia	0.40540014
7	Slovakia	0.39995460
8	Hungary	0.38580312
9	Czechia	0.38082003
10	Latvia	0.36522282
11	Finland	0.36268783
12	Luxembourg	0.35980871
13	Bulgaria	0.35711773
14	Croatia	0.35125471
15	Malta	0.35068126
16	Portugal	0.34860414
17	Cyprus	0.34768252
18	Lithuania	0.33747627
19	Greece	0.33541623
20	Poland	0.33487780
21	Germany	0.32908790
22	Sweden	0.32721403
23	Netherlands	0.31992851
24	France	0.30911838
25	Belgium	0.30789144
26	Italy	0.30781423
27	Spain	0.30187479

A.3 Axis 3: Technology

Table 22: Technology axis ranking, EU-27, 2024.

Rk	Country	Score
1	Ireland	0.58695939
2	Slovenia	0.35739912

Table 22: Technology axis ranking (continued).

Rk	Country	Score
3	Romania	0.29691153
4	Belgium	0.29454573
5	Austria	0.26057910
6	Lithuania	0.24172279
7	Estonia	0.22713142
8	Hungary	0.22538122
9	Cyprus	0.22438354
10	Slovakia	0.21649186
11	Poland	0.20845359
12	Malta	0.20546991
13	Luxembourg	0.20019721
14	Finland	0.19789321
15	Portugal	0.19766062
16	Spain	0.19263249
17	Czechia	0.18242269
18	Croatia	0.17343982
19	Greece	0.16272142
20	Bulgaria	0.15204272
21	Denmark	0.14995928
22	Italy	0.14103384
23	Latvia	0.13309607
24	Netherlands	0.12527375
25	France	0.12001059
26	Sweden	0.11982797
27	Germany	0.09762824

A.4 Axis 4: Defence

Table 23: Defence axis ranking, EU-27, 2024.

Rk	Country	Score
1	Malta	1.00000000
2	Netherlands	0.95955396
3	Italy	0.92057651
4	Sweden	0.88107943
5	Bulgaria	0.86064518

Table 23: Defence axis ranking (continued).

Rk	Country	Score
6	Cyprus	0.74847998
7	Croatia	0.74648317
8	Denmark	0.72368274
9	Slovenia	0.66661871
10	Luxembourg	0.64483831
11	Austria	0.58506379
12	Germany	0.58266751
13	Romania	0.56353195
14	Greece	0.55148593
15	Hungary	0.54471981
16	Czechia	0.48413737
17	Ireland	0.44887460
18	Belgium	0.44734854
19	Poland	0.44326493
20	Portugal	0.43376254
21	Lithuania	0.40227798
22	Finland	0.38964518
23	Spain	0.38088569
24	France	0.37034617
25	Estonia	0.35561474
26	Latvia	0.34079636
27	Slovakia	0.00000000

A.5 Axis 5: Critical Inputs

Table 24: Critical Inputs axis ranking, EU-27, 2024.

Rk	Country	Score
1	Finland	0.54699996
2	Croatia	0.52392772
3	Denmark	0.43629935
4	Cyprus	0.42359750
5	Malta	0.40874987
6	Estonia	0.38622776
7	Austria	0.31857781

Table 24: Critical Inputs axis ranking (continued).

Rk	Country	Score
8	Ireland	0.30719544
9	Portugal	0.25955617
10	Hungary	0.23405214
11	Germany	0.22312039
12	Slovenia	0.22301125
13	Luxembourg	0.22295831
14	Greece	0.22200621
15	Italy	0.20839534
16	Sweden	0.19257891
17	Belgium	0.17248330
18	Czechia	0.16976911
19	Poland	0.16532767
20	France	0.16122463
21	Romania	0.15814280
22	Slovakia	0.15490434
23	Latvia	0.15120012
24	Bulgaria	0.14843767
25	Lithuania	0.13056067
26	Netherlands	0.13018835
27	Spain	0.11235234

A.6 Axis 6: Logistics

Table 25: Logistics axis ranking, EU-27, 2024.

Rk	Country	Score
1	Malta	1.00000000
2	Cyprus	0.94106062
3	Finland	0.79733123
4	Sweden	0.75512656
5	Greece	0.71982808
6	Denmark	0.64230809
7	Luxembourg	0.61858285
8	Ireland	0.61373023
9	Czechia	0.52948100

Table 25: Logistics axis ranking (continued).

Rk	Country	Score
10	Austria	0.52592111
11	Portugal	0.50778629
12	Poland	0.48860717
13	Slovakia	0.48550648
14	Italy	0.47781001
15	Estonia	0.46987927
16	Hungary	0.46578765
17	Slovenia	0.44714621
18	Spain	0.43235135
19	Latvia	0.36822299
20	Bulgaria	0.36290606
21	Croatia	0.35798800
22	Netherlands	0.35491930
23	France	0.35293643
24	Belgium	0.35205038
25	Lithuania	0.34689716
26	Romania	0.28746783
27	Germany	0.27295297

A.7 Axis dominance detail

Table 26 lists the max-axis and second-max-axis for each country.

Table 26: Axis dominance per country: highest- and second-highest-scoring axes.

Rk	Country	Max Score	Max Axis	2nd Score	2nd Axis
1	Malta	1.00000000	Defence	1.00000000	Logistics
2	Cyprus	0.94106062	Logistics	0.74847998	Defence
3	Ireland	0.61373023	Logistics	0.58695939	Technology
4	Denmark	0.72368274	Defence	0.64230809	Logistics
5	Croatia	0.74648317	Defence	0.52392772	Critical Inputs
6	Finland	0.79733123	Logistics	0.54699996	Critical Inputs
7	Sweden	0.88107943	Defence	0.75512656	Logistics
8	Austria	0.58506379	Defence	0.52592111	Logistics
9	Italy	0.92057651	Defence	0.47781001	Logistics
10	Slovenia	0.66661871	Defence	0.44714621	Logistics

Table 26: Axis dominance per country (continued).

Rk	Country	Max Score	Max Axis	2nd Score	2nd Axis
11	Greece	0.71982808	Logistics	0.55148593	Defence
12	Luxembourg	0.64483831	Defence	0.61858285	Logistics
13	Estonia	0.46987927	Logistics	0.43935251	Energy
14	Bulgaria	0.86064518	Defence	0.36290606	Logistics
15	Netherlands	0.95955396	Defence	0.35491930	Logistics
16	Hungary	0.54471981	Defence	0.46578765	Logistics
17	Portugal	0.50778629	Logistics	0.43376254	Defence
18	Romania	0.56353195	Defence	0.40546401	Energy
19	Czechia	0.52948100	Logistics	0.48413737	Defence
20	Poland	0.48860717	Logistics	0.44326493	Defence
21	Belgium	0.44734854	Defence	0.35205038	Logistics
22	Germany	0.58266751	Defence	0.32908790	Energy
23	Lithuania	0.40227798	Defence	0.34689716	Logistics
24	Spain	0.43235135	Logistics	0.38088569	Defence
25	Latvia	0.36822299	Logistics	0.36522282	Energy
26	Slovakia	0.48550648	Logistics	0.39995460	Energy
27	France	0.37034617	Defence	0.35293643	Logistics

B Robustness Detail Tables

This appendix provides the full country-level detail for each robustness exercise summarised in Section 6.

B.1 Leave-one-axis-out: full rank table

Table 27 reports the rank of each country under each LOO variant (baseline, drop axis 1, . . . , drop axis 6), along with the maximum rank change from baseline.

Table 27: Detailed rank comparison: baseline vs. each leave-one-axis-out variant. Columns show the rank under each LOO composite. $\Delta_{\max}$ is the maximum absolute rank change across all six LOO variants.

Rk	**Country**	**−Fina**	**−Ener**	**−Tech**	**−Defe**	**−Crit**	**−Logi**	$\boldsymbol{\Delta_{\max}}$
1	Malta	1	1	1	2	1	1	1
2	Cyprus	2	2	2	3	2	5	3
3	Ireland	4	3	10	1	3	3	7
4	Denmark	3	4	3	5	5	4	1
5	Croatia	7	5	5	8	11	2	6
6	Finland	5	7	6	4	14	13	8
7	Sweden	6	6	4	12	4	11	5
8	Austria	8	10	8	6	8	7	2
9	Italy	10	8	7	19	6	8	10
10	Slovenia	9	11	14	10	7	6	4
11	Greece	12	9	9	9	9	17	6
12	Luxembourg	11	12	11	11	10	15	3
13	Estonia	15	15	15	7	18	14	6
14	Bulgaria	14	14	12	23	12	9	9
15	Netherlands	13	13	13	25	13	10	10
16	Hungary	16	16	16	14	16	16	2
17	Portugal	17	17	17	13	19	18	4
18	Romania	19	19	19	17	15	12	6
19	Czechia	18	18	18	15	17	19	4
20	Poland	20	20	20	18	20	22	2
21	Belgium	21	21	22	20	21	20	1
22	Germany	22	22	21	27	24	21	5
23	Lithuania	23	24	24	21	22	23	2
24	Spain	24	23	23	22	23	24	2
25	Latvia	25	25	25	24	25	25	1

Table 27: LOO detailed ranks (continued).

Rk	Country	−Fina	−Ener	−Tech	−Defe	−Crit	−Logi	$\Delta_{\max}$
26	Slovakia	27	27	27	16	26	27	10
27	France	26	26	26	26	27	26	1

B.2 Z-score standardisation: full rank comparison

Table 15 in Section 6.2 compares baseline and z-score ranks for all 27 countries. See Table 15 for the full table.

B.3 Dirichlet weight perturbation: full volatility table

Table 16 in Section 6.3 reports mean rank, rank standard deviation, and 5th/95th percentile ranks for all 27 countries. See Table 16 for the full table.

B.4 Winsorization: full rank comparison

Table 17 in Section 6.4 compares baseline and winsorized ranks for all 27 countries. See Table 17 for the full table.

C Reproducibility Protocol

This appendix documents the complete procedure for reproducing all results presented in this paper.

C.1 Software requirements

- **Python** $\geq$ 3.10 (tested with 3.14.0).
- **NumPy** $\geq$ 2.0 (tested with 2.2.6). NumPy is the only external dependency.
- **LaTeX** distribution with pdfLaTeX (tested with TeX Live 2024), including packages: `pgfplots` ($\geq$ 1.18), `tikz`, `siunitx`, `longtable`, `booktabs`, `natbib`, `cleveref`, `threeparttable`, `pdflscape`, `adjustbox`, `subcaption`, `listings`, `xcolor`.

C.2 Directory structure

The project is organised as follows:

```
paper2/
+-- main.tex                % Master document
+-- references.bib          % Bibliography
+-- sections/
|   +-- 00_executive_summary.tex
|   +-- 01_introduction.tex
|   +-- ...
|   +-- 10_data_availability.tex
+-- tables/                 % Generated .tex table fragments
|   +-- T06_axis_top_bottom.tex
|   +-- T07_axis_dominance.tex
|   +-- ...
|   +-- radar_*.dat
+-- figures/                % TikZ figure source files
|   +-- F01_histogram.tex
|   +-- ...
|   +-- F15_radar_FR.tex
+-- appendix/
|   +-- appendix_a_tables.tex
|   +-- appendix_b_robustness.tex
|   +-- appendix_c_reproducibility.tex
|   +-- appendix_d_code.tex
+-- scripts/
```

```
+-- isi_results_paper2.py
```

C.3 Step-by-step reproduction

1. **Install Python and NumPy.**

   ```
   python -m venv .venv
   source .venv/bin/activate
   pip install numpy
   ```

2. **Run the computation script.**

   ```
   cd paper2/scripts
   python isi_results_paper2.py
   ```

   This generates all `.tex` table fragments and `.dat` coordinate files in the `tables/` directory. The script embeds the full ISI v1.0 data snapshot and performs all computations (LOO, z-score, Dirichlet MC, winsorization, variance decomposition, Lorenz curve, ECDF, histograms, radar coordinates).

3. **Compile the LaTeX document.**

   ```
   cd paper2
   pdflatex main.tex
   bibtex main
   pdflatex main.tex
   pdflatex main.tex
   ```

   Three passes are required to resolve cross-references and the bibliography.

4. **Verify.** The final PDF should compile with 0 errors, 0 warnings (excluding font-related warnings), and 0 overfull boxes.

C.4 Random seed

The Dirichlet Monte Carlo exercise uses a fixed random seed (`np.random.seed(42)`) to ensure exact reproducibility of rank volatility statistics. Changing the seed will produce quantitatively similar but not identical results.

C.5 Data integrity

The ISI v1.0 data snapshot is embedded directly in the Python script (`isi_results_paper2.py`, lines 37–77) as a list of tuples. No external data files need to be downloaded or configured. The embedded data constitutes the single authoritative source for all computations in this paper.

D Computation Code

The complete Python script used to generate all derived tables and data files is listed below. The script requires only NumPy and embeds the full ISI v1.0 data snapshot.

Listing 1: ISI Results Paper 2 — Computation Engine (`isi_results_paper2.py`).

```python
#!/usr/bin/env python3
"""
ISI Results Paper 2 -- Computation Engine
=========================================
Reads the ISI v1.0 EU-27 snapshot (embedded) and produces all derived
tables for Paper 2:
  - Leave-one-axis-out (LOO) composites + Spearman correlations
  - Z-score standardized composites + rank comparison
  - Dirichlet weight perturbation (Monte Carlo) rank volatility
  - Winsorization (cap Axis 4 & 6 at p95) + rank comparison
  - Axis dominance table (max axis, 2nd max per country)
  - Variance contribution decomposition
  - Lorenz curve coordinates for composite
  - Top-5 / Bottom-5 per axis

All output is printed as LaTeX table fragments to stdout and also
written to ../tables/ as individual .tex files.

Usage:
    python isi_results_paper2.py

No external dependencies beyond numpy (ships with most scientific Python).
"""

import numpy as np
import os
import sys
from io import StringIO

# -- Authoritative ISI v1.0 Snapshot ----------------------------------------

AXIS_NAMES = [
    "Financial", "Energy", "Technology",
    "Defence", "Critical Inputs", "Logistics"
]
AXIS_SLUGS = [
    "financial", "energy", "technology",
    "defence", "critical_inputs", "logistics"
]

# fmt: off
DATA = [
    # (iso2, name, rank, composite, classification, a1, a2, a3, a4, a5, a6)
    ("MT", "Malta",        1,  0.51748148, "highly_concentrated",     0.13998786, 0.35068126, 0.20546991, 1.00000000, 0.40874987, 1.00000000),
    ("CY", "Cyprus",       2,  0.46821264, "moderately_concentrated", 0.12407171, 0.34768252, 0.22438354, 0.74847998, 0.42359750, 0.94106062),
    ("IE", "Ireland",      3,  0.42802519, "moderately_concentrated", 0.14666620, 0.46472525, 0.58695939, 0.44887460, 0.30719544, 0.61373023),
    ("DK", "Denmark",      4,  0.42442738, "moderately_concentrated", 0.12401663, 0.47029821, 0.14995928, 0.72368274, 0.43629935, 0.64230809),
    ("HR", "Croatia",      5,  0.40434768, "moderately_concentrated", 0.27299265, 0.35125471, 0.17343982, 0.74648317, 0.52392772, 0.35798800),
    ("FI", "Finland",      6,  0.40205136, "moderately_concentrated", 0.11775073, 0.36268783, 0.19789321, 0.38964518, 0.54699996, 0.79733123),
    ("SE", "Sweden",       7,  0.39871532, "moderately_concentrated", 0.11646504, 0.32721403, 0.11982797, 0.88107943, 0.19257891, 0.75512656),
    ("AT", "Austria",      8,  0.38249806, "moderately_concentrated", 0.14903467, 0.45581191, 0.26057910, 0.58506379, 0.31857781, 0.52592111),
    ("IT", "Italy",        9,  0.37386086, "moderately_concentrated", 0.18753526, 0.30781423, 0.14103384, 0.92057651, 0.20839534, 0.47781001),
    ("SI", "Slovenia",     10, 0.36979186, "moderately_concentrated", 0.11917574, 0.40540014, 0.35739912, 0.66661871, 0.22301125, 0.44714621),
    ("EL", "Greece",       11, 0.36537350, "moderately_concentrated", 0.20078311, 0.33541623, 0.16272142, 0.55148593, 0.22200621, 0.71982808),
    ("LU", "Luxembourg",   12, 0.36016893, "moderately_concentrated", 0.11462818, 0.35980871, 0.20019721, 0.64483831, 0.22295831, 0.61858285),
    ("EE", "Estonia",      13, 0.34332177, "moderately_concentrated", 0.18172493, 0.43935251, 0.22713142, 0.35561474, 0.38622776, 0.46987927),
    ("BG", "Bulgaria",     14, 0.34027984, "moderately_concentrated", 0.16052965, 0.35711773, 0.15204272,
```

```
        0.86064518, 0.14843767, 0.36290606),
    ("NL", "Netherlands",  15, 0.33515702, "moderately_concentrated", 0.12107826, 0.31992851, 0.12527375, 0.95955396, 0.13018835, 0.35491930),
    ("HU", "Hungary",      16, 0.32966227, "moderately_concentrated", 0.12222969, 0.38580312, 0.22538122, 0.54471981, 0.23405214, 0.46578765),
    ("PT", "Portugal",     17, 0.32173430, "moderately_concentrated", 0.18303606, 0.34860414, 0.19766062, 0.43376254, 0.25955617, 0.50778629),
    ("RO", "Romania",      18, 0.31864192, "moderately_concentrated", 0.20033340, 0.40546401, 0.29691153, 0.56353195, 0.15814280, 0.28746783),
    ("CZ", "Czechia",      19, 0.31796630, "moderately_concentrated", 0.16116762, 0.38082003, 0.18242269, 0.48413737, 0.16976911, 0.52948100),
    ("PL", "Poland",       20, 0.29561321, "moderately_concentrated", 0.13314812, 0.33487780, 0.20845359, 0.44326493, 0.16532767, 0.48860717),
    ("BE", "Belgium",      21, 0.28796781, "moderately_concentrated", 0.15348744, 0.30789144, 0.29454573, 0.44734854, 0.17248330, 0.35205038),
    ("DE", "Germany",      22, 0.26788694, "moderately_concentrated", 0.10186464, 0.32908790, 0.09762824, 0.58266751, 0.22312039, 0.27295297),
    ("LT", "Lithuania",    23, 0.26540520, "moderately_concentrated", 0.13349636, 0.33747627, 0.24172279, 0.40227798, 0.13056067, 0.34689716),
    ("ES", "Spain",        24, 0.26093977, "moderately_concentrated", 0.14554195, 0.30187479, 0.19263249, 0.38088569, 0.11235234, 0.43235135),
    ("LV", "Latvia",       25, 0.24695137, "mildly_concentrated",     0.12316988, 0.36522282, 0.13309607, 0.34079636, 0.15120012, 0.36822299),
    ("SK", "Slovakia",     26, 0.23641567, "mildly_concentrated",     0.16163673, 0.39995460, 0.21649186, 0.00000000, 0.15490434, 0.48550648),
    ("FR", "France",       27, 0.23567126, "mildly_concentrated",     0.10039135, 0.30911838, 0.12001059, 0.37034617, 0.16122463, 0.35293643),
]
# fmt: on

N = len(DATA)
assert N == 27

# Parse into arrays
ISO   = [d[0] for d in DATA]
NAMES = [d[1] for d in DATA]
RANKS = np.array([d[2] for d in DATA])
COMP  = np.array([d[3] for d in DATA])
CLASS = [d[4] for d in DATA]
AXES  = np.array([[d[5+j] for j in range(6)] for d in DATA])  # shape (27,6)

# Output directory
OUT_DIR = os.path.join(os.path.dirname(os.path.abspath(__file__)), "..", "tables")
os.makedirs(OUT_DIR, exist_ok=True)

def write_table(filename, content):
    path = os.path.join(OUT_DIR, filename)
    with open(path, "w") as f:
        f.write(content)
    print(f"  Written: {path}")

def rankdata(x):
    """Rank data (1 = highest value). Handles ties by averaging."""
    n = len(x)
    order = np.argsort(-x)  # descending
    ranks = np.empty(n, dtype=float)
    ranks[order] = np.arange(1, n+1, dtype=float)
    return ranks

def spearman_corr(r1, r2):
    """Spearman rank correlation between two rank arrays."""
    d = r1 - r2
    n = len(r1)
    return 1.0 - (6.0 * np.sum(d**2)) / (n * (n**2 - 1))


# ============================================================================
# 1. AXIS DOMINANCE TABLE
# ============================================================================
print("\n" + "="*72)
print("1. AXIS DOMINANCE")
print("="*72)

max_axis_idx = np.argmax(AXES, axis=1)
# Second max
second_max_idx = np.zeros(N, dtype=int)
for i in range(N):
    sorted_idx = np.argsort(-AXES[i])
    second_max_idx[i] = sorted_idx[1]

```

```python
# Count how many countries have each axis as max
axis_dom_counts = np.zeros(6, dtype=int)
for i in range(N):
    axis_dom_counts[max_axis_idx[i]] += 1

print("Axis dominance counts:")
for j in range(6):
    print(f"  {AXIS_NAMES[j]:20s}: {axis_dom_counts[j]}")

lines = []
lines.append("% Table T7: Axis dominance per country")
lines.append("\\begin{longtable}{@{}rlS[table-format=1.8]lS[table-format=1.8]l@{}}")
lines.append("\\caption{Axis dominance per country: highest- and second-highest-scoring axes.}")
lines.append("\\label{tab:axis-dominance}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Max Score}} & {\\textbf{Max Axis}} & {\\textbf{2nd Score}} & {\\textbf{2nd Axis}} \\\\")
lines.append("\\midrule")
lines.append("\\endfirsthead")
lines.append("\\caption[]{Axis dominance per country (continued).}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Max Score}} & {\\textbf{Max Axis}} & {\\textbf{2nd Score}} & {\\textbf{2nd Axis}} \\\\")
lines.append("\\midrule")
lines.append("\\endhead")
lines.append("\\bottomrule")
lines.append("\\endlastfoot")
for i in range(N):
    mi = max_axis_idx[i]
    si = second_max_idx[i]
    lines.append(f"  {RANKS[i]:.0f} & {NAMES[i]} & {AXES[i,mi]:.8f} & {AXIS_NAMES[mi]} & {AXES[i,si]:.8f} & {AXIS_NAMES[si]} \\\\")
lines.append("\\end{longtable}")
write_table("T07_axis_dominance.tex", "\n".join(lines))


# ============================================================================
# 2. VARIANCE CONTRIBUTION DECOMPOSITION
# ============================================================================
print("\n" + "="*72)
print("2. VARIANCE CONTRIBUTION")
print("="*72)

axis_vars = np.var(AXES, axis=0, ddof=0)  # population variance per axis
total_var_sum = np.sum(axis_vars)
# Covariance contribution
cov_matrix = np.cov(AXES.T, ddof=0)  # 6x6 population covariance
total_composite_var = np.var(COMP, ddof=0)
# Composite = mean of 6 axes, so Var(composite) = (1/36) * sum of all cov entries
var_decomp_check = np.sum(cov_matrix) / 36.0
print(f"  Composite variance (direct):    {total_composite_var:.10f}")
print(f"  Composite variance (from cov):  {var_decomp_check:.10f}")

# Marginal contribution of axis j = (1/36) * sum_k cov(j,k)
marginal_contrib = np.sum(cov_matrix, axis=1) / 36.0
pct_contrib = marginal_contrib / var_decomp_check * 100.0

lines = []
lines.append("% Table T8: Variance contribution decomposition")
lines.append("\\begin{table}[htbp]")
lines.append("\\centering")
lines.append("\\caption{Variance contribution decomposition of the ISI composite.  Each axis's")
lines.append("marginal contribution equals $(1/36)\\sum_{k=1}^{6}\\operatorname{Cov}(A_j, A_k)$,")
lines.append("reflecting both own-variance and covariance with other axes.}")
lines.append("\\label{tab:variance-decomp}")
lines.append("\\begin{tabular}{@{}lS[table-format=1.8]S[table-format=1.8]S[table-format=2.2]@{}}")
lines.append("\\toprule")
lines.append("{\\textbf{Axis}} & {\\textbf{Var (pop.)}} & {\\textbf{Marginal Contrib.}} & {\\textbf{\\% of Composite Var}} \\\\")
lines.append("\\midrule")
for j in range(6):
    lines.append(f"  {AXIS_NAMES[j]} & {axis_vars[j]:.8f} & {marginal_contrib[j]:.8f} & {pct_contrib[j]:.2f} \\\\")
lines.append("\\midrule")
lines.append(f"  Total & {total_var_sum:.8f} & {var_decomp_check:.8f} & {np.sum(pct_contrib):.2f} \\\\")
lines.append("\\bottomrule")
lines.append("\\end{tabular}")
lines.append("\\end{table}")
write_table("T08_variance_decomp.tex", "\n".join(lines))

print("Variance contribution:")
```

```python
for j in range(6):
    print(f"  {AXIS_NAMES[j]:20s}: var={axis_vars[j]:.8f}  marginal={marginal_contrib[j]:.8f}  pct={pct_contrib[j]:.2f}%")


# ========================================================================
# 3. LEAVE-ONE-AXIS-OUT (LOO)
# ========================================================================
print("\n" + "="*72)
print("3. LEAVE-ONE-AXIS-OUT")
print("="*72)

baseline_ranks = rankdata(COMP)

loo_composites = np.zeros((N, 6))
loo_ranks = np.zeros((N, 6))
loo_spearman = np.zeros(6)

for j in range(6):
    mask = [k for k in range(6) if k != j]
    loo_comp = np.mean(AXES[:, mask], axis=1)
    loo_composites[:, j] = loo_comp
    lr = rankdata(loo_comp)
    loo_ranks[:, j] = lr
    loo_spearman[j] = spearman_corr(baseline_ranks, lr)
    print(f"  Drop {AXIS_NAMES[j]:20s}: Spearman rho = {loo_spearman[j]:.6f}")

# Max rank change per country across all LOO
max_rank_change = np.zeros(N, dtype=int)
for i in range(N):
    changes = [abs(baseline_ranks[i] - loo_ranks[i, j]) for j in range(6)]
    max_rank_change[i] = int(max(changes))

# Table T9: LOO summary
lines = []
lines.append("% Table T9: Leave-one-axis-out rank changes")
lines.append("\\begin{table}[htbp]")
lines.append("\\centering")
lines.append("\\caption{Spearman rank correlation between baseline ISI ranking and each")
lines.append("leave-one-axis-out (LOO) ranking.  Higher $\\rho$ indicates lower sensitivity")
lines.append("to the excluded axis.}")
lines.append("\\label{tab:loo-summary}")
lines.append("\\begin{tabular}{@{}lS[table-format=1.6]@{}}")
lines.append("\\toprule")
lines.append("{\\textbf{Excluded Axis}} & {$\\boldsymbol{\\rho}$ \\textbf{(Spearman)}} \\\\")
lines.append("\\midrule")
for j in range(6):
    lines.append(f"  {AXIS_NAMES[j]} & {loo_spearman[j]:.6f} \\\\")
lines.append("\\bottomrule")
lines.append("\\end{tabular}")
lines.append("\\end{table}")
write_table("T09_loo_summary.tex", "\n".join(lines))

# Detailed LOO rank table (Appendix B)
lines = []
lines.append("% Appendix Table: LOO detailed ranks")
lines.append("\\begin{longtable}{@{}rl" + "r"*6 + "r@{}}")
lines.append("\\caption{Detailed rank comparison: baseline vs.\\ each leave-one-axis-out variant.")
lines.append("Columns show the rank under each LOO composite.  $\\Delta_{\\max}$ is the")
lines.append("maximum absolute rank change across all six LOO variants.}")
lines.append("\\label{tab:loo-detail}\\\\")
lines.append("\\toprule")
hdr = "{\\textbf{Rk}} & {\\textbf{Country}}"
for j in range(6):
    short = AXIS_SLUGS[j][:4].capitalize()
    hdr += f" & {{\\textbf{{$-${short}}}}}"
hdr += " & {$\\boldsymbol{\\Delta_{\\max}}$} \\\\"
lines.append(hdr)
lines.append("\\midrule")
lines.append("\\endfirsthead")
lines.append("\\caption[]{LOO detailed ranks (continued).}\\\\")
lines.append("\\toprule")
lines.append(hdr)
lines.append("\\midrule")
lines.append("\\endhead")
lines.append("\\bottomrule")
lines.append("\\endlastfoot")
for i in range(N):
    row = f"  {int(baseline_ranks[i])} & {NAMES[i]}"
    for j in range(6):
```

```
        row += f" & {int(loo_ranks[i,j])}"
    row += f" & {max_rank_change[i]} \\\\"
    lines.append(row)
lines.append("\\end{longtable}")
write_table("TB1_loo_detail.tex", "\n".join(lines))


# ============================================================================
# 4. Z-SCORE STANDARDIZED COMPOSITE
# ============================================================================
print("\n" + "="*72)
print("4. Z-SCORE STANDARDIZED COMPOSITE")
print("="*72)

axis_means = np.mean(AXES, axis=0)
axis_stds_pop = np.std(AXES, axis=0, ddof=0)
Z = (AXES - axis_means) / axis_stds_pop  # z-score each axis
z_composite = np.mean(Z, axis=1)
z_ranks = rankdata(z_composite)
z_spearman = spearman_corr(baseline_ranks, z_ranks)
print(f"  Spearman rho (baseline vs z-score): {z_spearman:.6f}")

lines = []
lines.append("% Table T10: Z-score standardized composite rank comparison")
lines.append("\\begin{longtable}{@{}rlS[table-format=1.8]rS[table-format=2.8]rr@{}}")
lines.append("\\caption{Rank comparison: baseline ISI vs. \\ z-score standardized composite.")
lines.append("The z-score composite re-centres each axis to mean zero and unit population")
lines.append("standard deviation before averaging.  Spearman $\\rho = " + f"{z_spearman:.6f}" + "$.}")
lines.append("\\label{tab:zscore-ranks}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Baseline}} & {\\textbf{Rk (base)}} & {\\textbf{Z-Composite}} & {\\textbf{Rk (z)}} & {$\\boldsymbol{\\Delta}$} \\\\")
lines.append("\\midrule")
lines.append("\\endfirsthead")
lines.append("\\caption[]{Z-score rank comparison (continued).}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Baseline}} & {\\textbf{Rk (base)}} & {\\textbf{Z-Composite}} & {\\textbf{Rk (z)}} & {$\\boldsymbol{\\Delta}$} \\\\")
lines.append("\\midrule")
lines.append("\\endhead")
lines.append("\\bottomrule")
lines.append("\\endlastfoot")
# Sort by baseline rank
order = np.argsort(baseline_ranks)
for i in order:
    delta = int(abs(baseline_ranks[i] - z_ranks[i]))
    lines.append(f"  {int(baseline_ranks[i])} & {NAMES[i]} & {COMP[i]:.8f} & {int(baseline_ranks[i])} & {z_composite[i]:.8f} & {int(z_ranks[i])} & {delta} \\\\")
lines.append("\\end{longtable}")
write_table("T10_zscore_ranks.tex", "\n".join(lines))


# ============================================================================
# 5. DIRICHLET WEIGHT PERTURBATION (MONTE CARLO)
# ============================================================================
print("\n" + "="*72)
print("5. DIRICHLET WEIGHT PERTURBATION")
print("="*72)

np.random.seed(20240222)
N_MC = 10000
alpha = np.ones(6) * 10.0  # Concentrated around equal weights (1/6 each)

mc_ranks = np.zeros((N, N_MC))
for t in range(N_MC):
    w = np.random.dirichlet(alpha)
    mc_comp = AXES @ w
    mc_ranks[:, t] = rankdata(mc_comp)

# Summary statistics per country
mean_rank = np.mean(mc_ranks, axis=1)
std_rank = np.std(mc_ranks, axis=1, ddof=0)
p5_rank = np.percentile(mc_ranks, 5, axis=1)
p95_rank = np.percentile(mc_ranks, 95, axis=1)
prob_top5 = np.mean(mc_ranks <= 5, axis=1) * 100.0
prob_top10 = np.mean(mc_ranks <= 10, axis=1) * 100.0

lines = []
lines.append("% Table T11: Dirichlet weight perturbation rank volatility")
lines.append("\\begin{longtable}{@{}rl S[table-format=2.2] S[table-format=1.2] S[table-format=2.1] S[table-format
```

```
=2.1] S[table-format=3.1] S[table-format=3.1]@{}}")
lines.append("\\caption{Rank volatility under Dirichlet weight perturbation ($\\alpha_j=10$, $N_{\\text{MC}}=" + str(N_MC) + "$).")
lines.append("Mean rank, standard deviation, 5th--95th percentile band, and probability of")
lines.append("falling in the top-5 or top-10 are reported for each country.}")
lines.append("\\label{tab:dirichlet-volatility}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Mean Rk}} & {$\\boldsymbol{\\sigma}$} & {\\textbf{P5}} & {\\textbf{P95}} & {\\textbf{\\%Top5}} & {\\textbf{\\%Top10}} \\\\")
lines.append("\\midrule")
lines.append("\\endfirsthead")
lines.append("\\caption[]{Dirichlet rank volatility (continued).}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Mean Rk}} & {$\\boldsymbol{\\sigma}$} & {\\textbf{P5}} & {\\textbf{P95}} & {\\textbf{\\%Top5}} & {\\textbf{\\%Top10}} \\\\")
lines.append("\\midrule")
lines.append("\\endhead")
lines.append("\\bottomrule")
lines.append("\\endlastfoot")
order = np.argsort(baseline_ranks)
for i in order:
    lines.append(f"  {int(baseline_ranks[i])} & {NAMES[i]} & {mean_rank[i]:.2f} & {std_rank[i]:.2f} & {p5_rank[i]:.1f} & {p95_rank[i]:.1f} & {prob_top5[i]:.1f} & {prob_top10[i]:.1f} \\\\")
lines.append("\\end{longtable}")
write_table("T11_dirichlet_volatility.tex", "\n".join(lines))

# Also compute mean Spearman across MC draws
mc_spearmans = np.array([spearman_corr(baseline_ranks, mc_ranks[:, t]) for t in range(N_MC)])
print(f"  Mean Spearman rho across {N_MC} MC draws: {np.mean(mc_spearmans):.6f}")
print(f"  Std  Spearman rho:                        {np.std(mc_spearmans):.6f}")
print(f"  Min  Spearman rho:                        {np.min(mc_spearmans):.6f}")
print(f"  Max  Spearman rho:                        {np.max(mc_spearmans):.6f}")

# Countries with highest rank volatility
print("\n  Top 5 most volatile:")
vol_order = np.argsort(-std_rank)
for idx in vol_order[:5]:
    print(f"    {NAMES[idx]:15s}: mean_rk={mean_rank[idx]:.2f} std={std_rank[idx]:.2f} band=[{p5_rank[idx]:.0f},{p95_rank[idx]:.0f}]")


# ============================================================================
# 6. WINSORIZATION (CAP AXIS 4 & 6 AT P95)
# ============================================================================
print("\n" + "="*72)
print("6. WINSORIZATION")
print("="*72)

AXES_W = AXES.copy()
for j_cap in [3, 5]:  # Axis 4 (index 3) and Axis 6 (index 5)
    p95 = np.percentile(AXES[:, j_cap], 95)
    print(f"  {AXIS_NAMES[j_cap]} P95 cap: {p95:.8f}")
    AXES_W[:, j_cap] = np.minimum(AXES[:, j_cap], p95)

w_composite = np.mean(AXES_W, axis=1)
w_ranks = rankdata(w_composite)
w_spearman = spearman_corr(baseline_ranks, w_ranks)
print(f"  Spearman rho (baseline vs winsorized): {w_spearman:.6f}")

lines = []
lines.append("% Table T12: Winsorization sensitivity rank changes")
lines.append("\\begin{longtable}{@{}rlS[table-format=1.8]rS[table-format=1.8]rr@{}}")
lines.append("\\caption{Rank comparison: baseline vs.\\ winsorized composite (Axis-4 and Axis-6")
lines.append("capped at their respective 95th percentiles).  Spearman $\\rho = " + f"{w_spearman:.6f}" + "$.}")
lines.append("\\label{tab:winsorization}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Baseline}} & {\\textbf{Rk (base)}} & {\\textbf{Winsorized}} & {\\textbf{Rk (win)}} & {$\\boldsymbol{\\Delta}$} \\\\")
lines.append("\\midrule")
lines.append("\\endfirsthead")
lines.append("\\caption[]{Winsorization rank changes (continued).}\\\\")
lines.append("\\toprule")
lines.append("{\\textbf{Rk}} & {\\textbf{Country}} & {\\textbf{Baseline}} & {\\textbf{Rk (base)}} & {\\textbf{Winsorized}} & {\\textbf{Rk (win)}} & {$\\boldsymbol{\\Delta}$} \\\\")
lines.append("\\midrule")
lines.append("\\endhead")
lines.append("\\bottomrule")
lines.append("\\endlastfoot")
order = np.argsort(baseline_ranks)
for i in order:
```

```python
    delta = int(abs(baseline_ranks[i] - w_ranks[i]))
    lines.append(f"  {int(baseline_ranks[i])} & {NAMES[i]} & {COMP[i]:.8f} & {int(baseline_ranks[i])} & {w_composite[i]:.8f} & {int(w_ranks[i])} & {delta} \\\\")
lines.append("\\end{longtable}")
write_table("T12_winsorization.tex", "\n".join(lines))


# ============================================================================
# 7. TOP-5 / BOTTOM-5 PER AXIS
# ============================================================================
print("\n" + "="*72)
print("7. TOP-5 / BOTTOM-5 PER AXIS")
print("="*72)

lines = []
lines.append("% Table T6: Top-5 and Bottom-5 per axis")
lines.append("\\begin{table}[htbp]")
lines.append("\\centering")
lines.append("\\caption{Top-5 (most concentrated) and Bottom-5 (least concentrated) countries per axis.}")
lines.append("\\label{tab:axis-top-bottom}")
lines.append("\\footnotesize")
lines.append("\\begin{tabular}{@{}lllll@{}}")
lines.append("\\toprule")
lines.append("{\\textbf{Axis}} & {\\textbf{Top-5 (most conc.)}} & {\\textbf{Score}} & {\\textbf{Bottom-5 (least conc.)}} & {\\textbf{Score}} \\\\")
lines.append("\\midrule")

for j in range(6):
    sorted_idx = np.argsort(-AXES[:, j])
    top5 = sorted_idx[:5]
    bot5 = sorted_idx[-5:][::-1]  # reversed so worst is last
    for k in range(5):
        prefix = AXIS_NAMES[j] if k == 0 else ""
        ti = top5[k]
        bi = bot5[k]
        lines.append(f"  {prefix} & {NAMES[ti]} & {AXES[ti,j]:.8f} & {NAMES[bi]} & {AXES[bi,j]:.8f} \\\\")
    if j < 5:
        lines.append("\\addlinespace")

lines.append("\\bottomrule")
lines.append("\\end{tabular}")
lines.append("\\end{table}")
write_table("T06_axis_top_bottom.tex", "\n".join(lines))


# ============================================================================
# 8. LORENZ CURVE COORDINATES
# ============================================================================
print("\n" + "="*72)
print("8. LORENZ CURVE")
print("="*72)

sorted_comp = np.sort(COMP)
cum_share = np.cumsum(sorted_comp) / np.sum(sorted_comp)
pop_share = np.arange(1, N+1) / N
gini = 1.0 - 2.0 * np.trapezoid(cum_share, pop_share)
print(f"  Gini coefficient of composite: {gini:.6f}")

# Write coordinates for pgfplots
lines = []
lines.append("% Lorenz curve coordinates for pgfplots")
lines.append("% Format: population_share cumulative_score_share")
lines.append("0.0000 0.0000")
for k in range(N):
    lines.append(f"{pop_share[k]:.6f} {cum_share[k]:.6f}")
write_table("lorenz_coords.dat", "\n".join(lines))


# ============================================================================
# 9. AXIS RANK TABLES (for Appendix A)
# ============================================================================
print("\n" + "="*72)
print("9. AXIS RANK TABLES")
print("="*72)

for j in range(6):
    ax_ranks = rankdata(AXES[:, j])
    sorted_idx = np.argsort(ax_ranks)
    lines = []
    lines.append(f"% Appendix: {AXIS_NAMES[j]} axis ranking")
```

```
    lines.append("\\begin{longtable}{@{}rlS[table-format=1.8]@{}}")
    lines.append(f"\\caption{{{AXIS_NAMES[j]}␣axis␣ranking,␣EU-27,␣2024.}}")
    lines.append(f"\\label{{tab:axis-rank-{AXIS_SLUGS[j]}}}\\\\")
    lines.append("\\toprule")
    lines.append("{\\textbf{Rk}}␣&␣{\\textbf{Country}}␣&␣{\\textbf{Score}}␣\\\\")
    lines.append("\\midrule")
    lines.append("\\endfirsthead")
    lines.append(f"\\caption[]{{{AXIS_NAMES[j]}␣axis␣ranking␣(continued).}}\\\\")
    lines.append("\\toprule")
    lines.append("{\\textbf{Rk}}␣&␣{\\textbf{Country}}␣&␣{\\textbf{Score}}␣\\\\")
    lines.append("\\midrule")
    lines.append("\\endhead")
    lines.append("\\bottomrule")
    lines.append("\\endlastfoot")
    for i in sorted_idx:
        lines.append(f"␣␣{int(ax_ranks[i])}␣&␣{NAMES[i]}␣&␣{AXES[i,j]:.8f}␣\\\\")
    lines.append("\\end{longtable}")
    write_table(f"TA_{AXIS_SLUGS[j]}_ranking.tex", "\n".join(lines))
    print(f"␣␣Written␣axis␣ranking:␣{AXIS_NAMES[j]}")


# ============================================================================
# 10. HISTOGRAM COORDINATES FOR PGFPLOTS
# ============================================================================
print("\n" + "="*72)
print("10.␣HISTOGRAM/ECDF␣COORDINATES")
print("="*72)

# Composite values sorted for ECDF
sorted_comp = np.sort(COMP)
lines = []
lines.append("%␣ECDF␣coordinates")
lines.append("%␣Format:␣composite_score␣ecdf_value")
for k in range(N):
    lines.append(f"{sorted_comp[k]:.8f}␣{(k+1)/N:.8f}")
write_table("ecdf_coords.dat", "\n".join(lines))

# Histogram bin counts (10 bins)
bin_edges = np.linspace(0.20, 0.55, 8)
hist_counts, _ = np.histogram(COMP, bins=bin_edges)
lines = []
lines.append("%␣Histogram␣coordinates␣(bin_center,␣count)")
for k in range(len(hist_counts)):
    center = (bin_edges[k] + bin_edges[k+1]) / 2.0
    lines.append(f"{center:.4f}␣{hist_counts[k]}")
write_table("histogram_coords.dat", "\n".join(lines))


# ============================================================================
# 11. RADAR PLOT DATA PER COUNTRY (for TikZ)
# ============================================================================
print("\n" + "="*72)
print("11.␣RADAR␣PLOT␣DATA")
print("="*72)

selected = ["MT", "CY", "IE", "NL", "DE", "SK", "FI", "FR"]
for iso in selected:
    idx = ISO.index(iso)
    lines = []
    lines.append(f"%␣Radar␣data␣for␣{NAMES[idx]}␣({iso})")
    lines.append("%␣axis_index␣axis_name␣score")
    for j in range(6):
        lines.append(f"{j}␣{AXIS_SLUGS[j]}␣{AXES[idx,j]:.8f}")
    write_table(f"radar_{iso}.dat", "\n".join(lines))
    print(f"␣␣{NAMES[idx]}:␣{AXES[idx,:]}")


# ============================================================================
# 12. BOXPLOT STATS PER AXIS (for pgfplots)
# ============================================================================
print("\n" + "="*72)
print("12.␣BOXPLOT␣STATISTICS")
print("="*72)

for j in range(6):
    vals = AXES[:, j]
    q1 = np.percentile(vals, 25)
    med = np.median(vals)
    q3 = np.percentile(vals, 75)
    iqr = q3 - q1
```

```
    whisker_lo = np.min(vals[vals >= q1 - 1.5*iqr])
    whisker_hi = np.max(vals[vals <= q3 + 1.5*iqr])
    outliers_lo = vals[vals < q1 - 1.5*iqr]
    outliers_hi = vals[vals > q3 + 1.5*iqr]
    print(f"  {AXIS_NAMES[j]:20s}: Q1={q1:.4f} Med={med:.4f} Q3={q3:.4f} Lo={whisker_lo:.4f} Hi={whisker_hi:.4f}
         Out_lo={len(outliers_lo)} Out_hi={len(outliers_hi)}")


# ========================================================================
# DONE
# ========================================================================
print("\n" + "="*72)
print("ALL COMPUTATIONS COMPLETE")
print(f"Output tables written to: {OUT_DIR}")
print("="*72)
```